Nittobo

2-4-1, Kojimachi, Chiyoda-ku, Tokyo, 102-8489, Japan
Telephone: +81-3-4582-5111
https://www.nittobo.co.jp/eng/

Nittobo

Nittobo Group Integrated Report 2025



The Nittobo Group's Philosophy System

MISSION

Corporate Philosophy

The Nittobo Group will enhance its value to society as a corporate group by helping to create healthy and more comfortable lifestyles, and in so doing continue contributing to the realization of a more prosperous world.

VALUES

Nittobo Declaration

The Nittobo Group aims to be the "Best Partner" for society.
(Nittobo BP Declaration)

We relentlessly strive to identify our customers' needs and find great satisfaction in steadfastly earning their confidence and trust. In addition, we attach great importance to sharing this satisfaction with all of our stakeholders and society as a whole, including shareholders, investors, local governments and regional communities, through the provision of our products and services.

We are committed to being a corporate group that values the potential of all employees as individuals and inspires them to freely and willingly share and discuss their ideas, and ultimately achieving greater results through teamwork.

Our corporate group provides employees with opportunities for progress and self-realization based on the belief that their progress leads to everyone's success. We encourage our employees to be good citizens, think deeply, observe widely, act courageously and approach their jobs with resolve and determination.



Deeply, Widely, Strongly and Hospitably

Proud that customers trust Nittobo

VISION

Nittobo Group's Vision for 2030

Big VISION 2030

As a corporate group that continues to create Global No. 1 Niche Businesses, we aspire to contribute in the areas of Environment & Energy, Digital Society, and Health, Security & Safety with the goal of building a more sustainable society.

- Provide products and services that contribute to the environment and energy
- Provide products and services that contribute to the digital society
- Provide products and services that contribute to health, security and safety

We aspire to be a company that gains the trust of all stakeholders in the pursuit of client satisfaction.

CONTENTS

Section 1: Introduction
How Nittobo Contributes to Society 02
Message from the Chairman 04
Message from the CEO 05

Section 2: Value Creation Story
History of Value Creation 10
Value Creation Process 12
Value Creation Drivers and Capital 14
Nittobo Group's Material Issues 16
Big VISION 2030 18
Medium-Term Management Plan (FY2024–2027) 19
Message from the Division General Manager of the Corporate Management Division 21
Message from the Division General Manager of the Technology Development Division 22

Section 3: Strategies by Business
Business Overview 24
Electronic Materials Business 25
Medical Business 26
Composite Materials Business 27
Materials Solution Business 28
Insulation Materials Business 29
Nittobo's Special Glass: Pioneering the Future of a Society Characterized by High-Speed, High-Capacity Communications 30

Section 4: Sustainability Management
Nittobo's Sustainability Management 32
Environmental Initiatives 34
Active Roles for Diverse Talent 39
Safety Initiatives 44
Basic Stance on Human Rights / Sustainable Procurement 45
Directors and Executive Officers 46
Message from External Directors 48
Corporate Governance 52
Compliance and Risk Management 60

Section 5: Corporate Data
Financial and Non-Financial Highlights 61
11-Year Financial Data 64
Company Profile 66
Stock Information 67

Editorial Policy

The Nittobo Group Integrated Report summarizes the Nittobo Group's framework and core elements for value creation, and brings together financial and non-financial information and medium-to-long-term strategies in a single publication. When editing Nittobo Group Integrated Report 2025, which covers the progress of the four-year Medium-Term Management Plan launched in FY2024, we set out to strengthen our communication of how the Nittobo Group helps resolve pressing social issues through its business activities, offer additional content discussing sustainability management as the foundation that underpins our value creation, and enhance dialogue with stakeholders.

Report Period

This report covers business activities in FY2024 (from April 1, 2024 through March 31, 2025).
Note: Charts, graphs and tables may use the abbreviation "FY24."

Scope of Report

This report covers Nitto Boseki Co., Ltd., as well as its domestic and overseas subsidiaries and affiliated companies.

Date of Publication

September 30, 2025

Company References

"Nittobo Group" and "the Group" refer to Nitto Boseki Co., Ltd. and its consolidated subsidiaries and affiliates in Japan and overseas.
"Nittobo" and "the Company" refer to Nitto Boseki Co., Ltd. on a non-consolidated basis.

Forward-looking Statements

The forecasts presented in this report are based on information available to the Company as of the publication date.
Forward-looking statements included in this report do not represent promises or guarantees that the Company will attain its forecasts or realize the business measures expressed herein.

How Nittobo Contributes to Society

Products and services that contribute to the environment and energy

Materials that contribute to reducing environmental impact through energy conservation and fuel efficiency

Lightweight and strong, glass fiber is used in automobile and aircraft parts to make them lighter. Glass wool is used for insulation in homes and other applications, and contributes to energy efficiency by keeping indoor temperatures steady.

Products and services that contribute to the digital society

Helping bring about an advanced digital society that offers more comfortable and convenient lifestyles

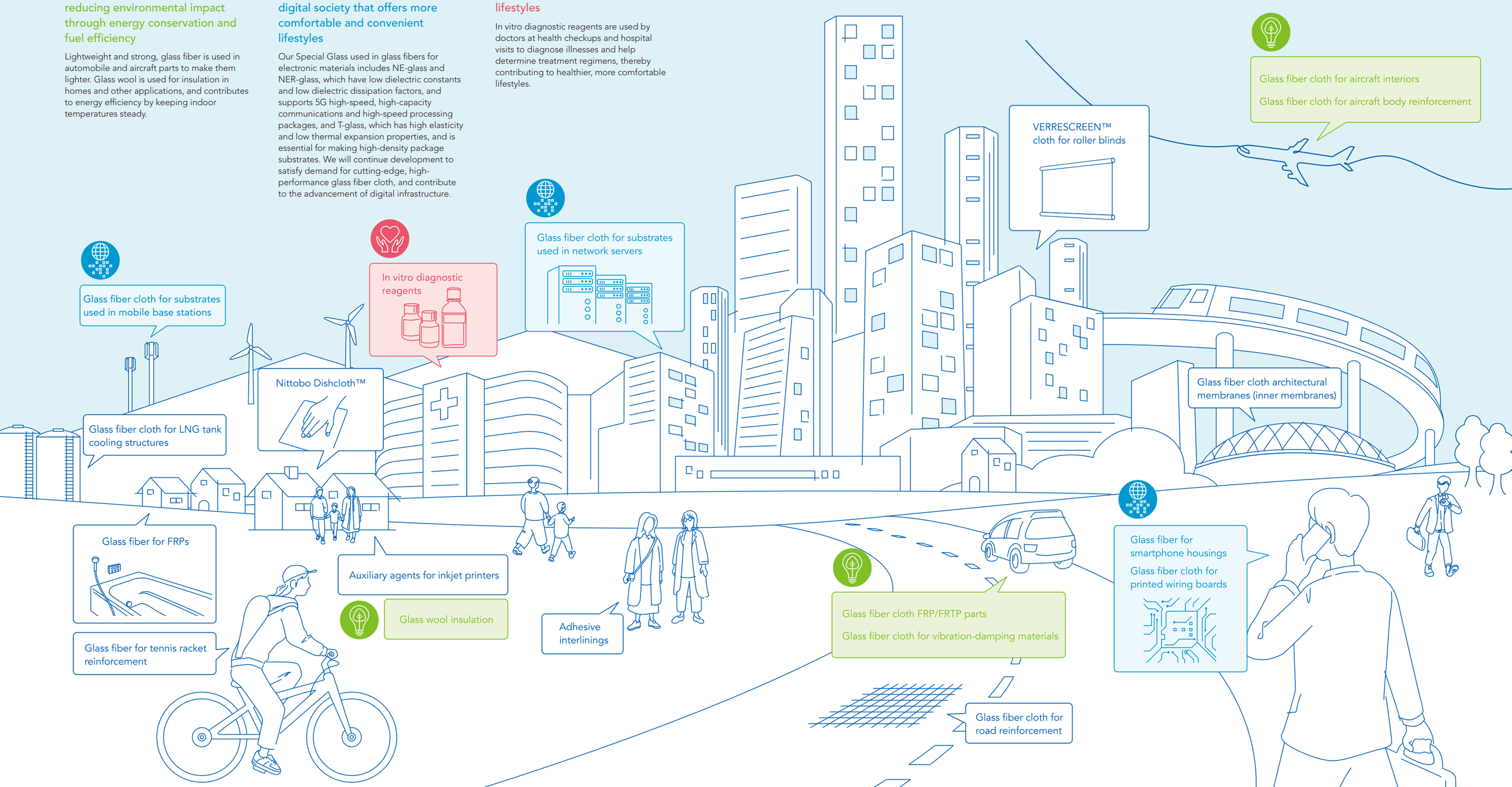
Our Special Glass used in glass fibers for electronic materials includes NE-glass and NER-glass, which have low dielectric constants and low dielectric dissipation factors, and supports 5G high-speed, high-capacity communications and high-speed processing packages, and T-glass, which has high elasticity and low thermal expansion properties, and is essential for making high-density package substrates. We will continue development to satisfy demand for cutting-edge, high-performance glass fiber cloth, and contribute to the advancement of digital infrastructure.

Products and services that contribute to health, security and safety

Products that support healthier lifestyles

In vitro diagnostic reagents are used by doctors at health checkups and hospital visits to diagnose illnesses and help determine treatment regimens, thereby contributing to healthier, more comfortable lifestyles.

Out of our earnest desire to make society and the future more convenient and enriched, the Nittobo Group has kept creating novel materials and technologies with a spirit of challenge that does not fear failure. We want to keep supporting a convenient and reassuring society; we want to make the future exciting. For this purpose, we will go on creating things the world has never known before.



Message from the Chairman

The feelings behind the words, “Proud that customers trust Nittobo.”

Yuichi Tsuji
Director, Representative and Chairman



The Nittobo Group has set itself the goal of being the “Best Partner” for society by pursuing the corporate philosophy of “striving to create healthy and comfortable lifestyles.” From the perspective of our stakeholders, this can be rephrased as “aspiring to be a company that gains the trust of all stakeholders in the pursuit of client satisfaction.” For that reason, what I considered most important when I became president nine years ago was to ensure that Nittobo Group continues to be a company where employees feel fortunate to be a part of. Only by being a company that continues to foster an open and communicative corporate culture in which employees remember to respect one another, work freely and comfortably, and take on challenges, can we create excellent products and services. If we can achieve this, we will gain and maintain the trust of all stakeholders, including our customers, suppliers, regional communities, and shareholders, in the pursuit of client satisfaction.

Throughout our more than 100 years of history, the Nittobo Group has developed and launched many products to meet the needs of society. This is our founding spirit: “Let’s turn anything into fibers.” Through our many failures and successes, we have always anticipated the future and repeatedly challenged ourselves to develop new products and services. As we embark on a new century, it is important to return to our culture of taking on challenges and must promote activities that lay the foundations for future success, developing long-term perspectives amid ever-changing social issues and market needs.

I hope that all our stakeholders will continue to offer their generous understanding and support.

Message from the CEO

Toward a new stage of growth: Expanding the scale of our business and developing new businesses

Hiroyuki Tada
Director, Representative and Chief Executive Officer

Furthering Investments and Cutting-Edge Winning Strategies

Looking back on FY2024, operating profit reached JPY 16.4 billion, exceeding the originally planned JPY 12.5 billion. The primary driver of profit was steady sales in Special Glass within the Electronic Materials Business. Sales of Special Glass had been somewhat sluggish through FY2022, but strong demand, driven by the growing AI server market and other factors, contributed significantly to both net sales and operating profit. We are currently expanding production capacity to meet this demand.

At present, market demand seems to have increased by about two to three times compared with when the Medium-Term Management Plan (FY2024–2027) was formulated. Since Chinese and Taiwanese companies are aggressively catching up with us in the Special Glass sector, it is crucial to make appropriate investment decisions. In terms of strategies by product, we have an overwhelming competitive advantage in both quality and production capacity for low thermal expansion glass and will continue to maintain this advantage going forward. As the transition from first-generation NE to NER and NEZ low-dielectric glass accelerates, the Company is currently conducting research and development on the fourth generation (DXII). To achieve this, while continuing to compete with our advanced technological capabilities, we are implementing a strategy of winning at the cutting edge, including the selection and elimination of products for commodity use. As we make our next investments, we are also strategically considering pricing to enhance profits.



Message from the CEO

Steady Implementation of the Medium-Term Management Plan, and Creation of New Businesses for the Next 100 Years

When we were formulating the Medium-Term Management Plan, we planned to make approximately JPY 80 billion in capital expenditures over four years, and we have been expanding our production facilities throughout FY2025. As part of this effort, in August of this year, we decided to expand our new plant building to strengthen our glass fiber cloth production system. As mentioned earlier, demand for electronic materials is exceeding expectations, so we are considering redesigning our investment plans. If we want to expand our production capacity for FY2028 to FY2029, the investment decision must be made either this year or next year. Moving forward, we will continue to expand our facilities with a sense of urgency.

The Company marked Nittobo's centenary in 2023, but the creation of new pillars for the next 100 years has yet to take shape and is still only at the stage of newly germinated seeds. It takes a long period of time to nurture a single business. For example, our Medical Business took 40 years to reach its current scale. It has been over 20 years since we developed Special Glass. And it has now finally become a pillar of profit. Those next pillars, the businesses that will become the next pillars, are still seeds, and we are continuing our exploration not to overlook them.

The New Business Development Center, established within the Company in April 2023, is leading efforts to create such new businesses. To pioneer a new future, we must emphasize free innovation activities through co-creation with external partners and cross-sector exchanges. As a specific example, we began a joint research project with Nippon Veterinary and Life Science University in 2025.

The Fukuyama Enterprise Center also promotes various collaborative initiatives between industry and academia, primarily through joint research with Tohoku University.

Even in existing businesses such as glass fiber cloth, possibilities remain. Amidst their busy work schedules, I hope our employees will continue to discover new ideas through conversations with various customers and researchers.

Effects of the Organizational Structure Restructuring

In terms of organizational structure, we reorganized our structure from the previous three business divisions to five business divisions in April 2024. This was done to strengthen activities that consider the customer's perspective by creating a new pillar following Special Glass and the medical field, while also integrating development, manufacturing, and sales operations. This restructuring of the organizational structure has already proven successful in various settings.

First, reorganizing into a business division system has significantly improved the line of communication across all levels. The Division General Managers of each Business Division are accessible to employees, so even frontline staff can now speak freely and directly in meetings.

Another success is that, regarding strengthening the consideration of customer's perspective, market-in thinking has permeated throughout the organization, from the heads of each Business Division down to the front lines. The Electronic Materials Business Division strives to accurately grasp market needs by continuously engaging in dialogue with primary, secondary, and end users. This has had a positive effect, such as significantly deepening the interaction between the head of the Business Divisions and our customers.

In all Business Divisions, the quality and quantity of technical meetings involving both our customers' development teams and the Company's development teams have improved. This has strengthened our structure for conducting thorough assessments and development based on the information obtained in these meetings. Specifically, the Materials Solution Business Division is gathering information on new innovations and exploring future pillars for the next 20 to 30 years, and our expectations are high. The Composite Materials Business Division had a sense of crisis over its past reliance on the market and is now generating ideas to transform the market itself.

The Electronic Materials Business Division is also seeing innovative ideas emerging that are focused on the market. In the field of electronic materials, which we have positioned as a main business, the pace of technological innovation is accelerating and we feel the need to respond with a sense of urgency based on market feedback.

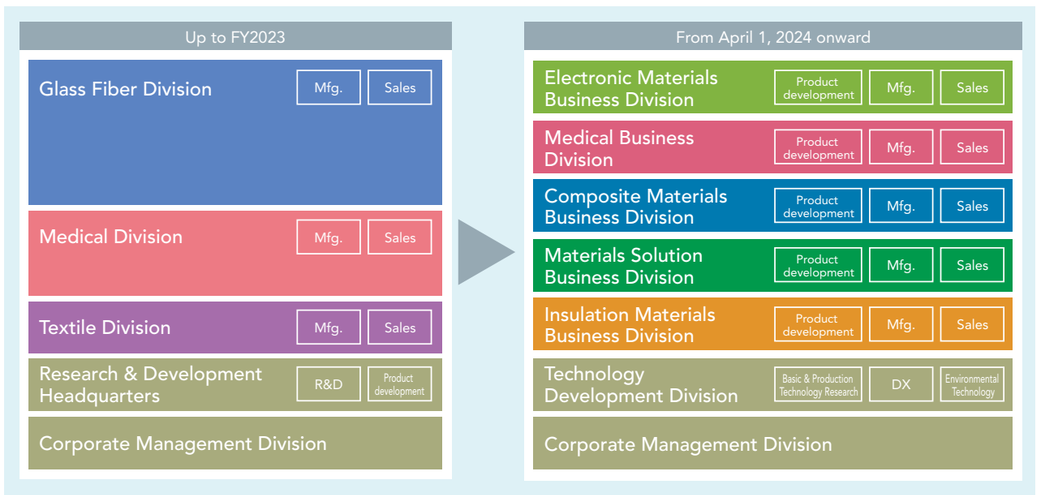
Since the reorganization of our structure into five business divisions, each business division has increasingly become a mini-company, and the distinct personalities of the Division General Managers have also become more pronounced. This trend is also very effective from the perspective of succession planning. The Company has changed its organization to be a company with a nomination committee, etc., and this trend has also been positively recognized by the nomination committee itself.

Employees' Ambition for Success will Build the Next Pillar of Nittobo

In sales, there are many young employees in their 20s and 30s, while the middle management layer tends to be thin. This is an issue for human resource development. I would like our young employees to act with a clear vision of what their work might look like when they are in their 40s and 50s. I think work is not enjoyable unless one is ambitious. When people decide to give it their all, that's when the next pillar of Nittobo will be born. From that perspective, I want to see even more ambition in our employees. The Company has a very high rate of paid



Restructuring of Business Segments



Message from the CEO

leave taken, and male employees also actively take paternity leave. We want employees to devote themselves to their work while effectively utilizing these systems.

Starting in FY2024, we launched a system to pay an additional third bonus (a special bonus linked to the Medium-Term Management Plan) when operating profit targets are achieved on a single-year basis. Securing human resources is one of the Company's greatest challenges, and we feel the third bonus has been effective in boosting employee motivation.

Securing human resources for our production base in Fukushima Prefecture is also a major challenge in our human capital recruitment efforts. We recognize that Fukushima Prefecture has a high outflow of young people and faces structural issues. We must firmly cultivate human resources that possess expertise from the front lines of manufacturing. By communicating the appeal of Nittobo even more powerfully, we can make it widely known to young people that there are opportunities for them to thrive here.

Making Large-Scale Investments While Maintaining Financial Soundness

The Medium-Term Management Plan promotes management practices that are conscious of ROE, ROIC, and the cost of capital. By ramping up implementation of our growth strategy, we aim to achieve a level of ROE that consistently exceeds the cost of capital. ROE for FY2024 was 10.4%. We were also able to secure ROIC in excess of WACC.

We have committed to a minimum dividend of JPY 55 per share. For FY2024, the interim dividend was JPY 27.50, the year-end dividend was JPY 78.50, and the annual dividend was JPY 106.00, resulting in a payout ratio of 30.1%.

The essence of financial capital policy lies in how to realize large-scale investments while maintaining financial soundness. As cross-holdings continue to be unwound, it is my mission to reach out to investors who will hold the Company's shares for the long term. While our investor relations activities for overseas investors have primarily focused on Hong Kong and Singapore so far, we will expand our IR activities to include investors in Europe, the Middle East, and other regions who will hold our shares long-term.

Continuing Initiatives on Key Issues in Sustainability Management Through Dialogue with Local Governments and Other Companies

In our long-term vision, Big VISION 2030, we have set out to contribute to society in three areas: the environment and energy; digital society; and health, security, and safety, while addressing ESG issues as a core theme. In our sustainability initiatives, we are continuously advancing research and development related to recycled glass. In our Insulation Materials Business, we utilize recycled glass for the majority of our raw materials. We will also pursue opportunities in the Composite Materials Business going forward.

As I mentioned earlier, increasing production capacity is an urgent issue. We have committed to reducing CO_2 emissions by 30% (compared to FY2013 levels) by 2030, and with this target serving as a milestone, we are ramping up our shift to electric furnaces. Initiatives toward achieving carbon neutrality by 2050 and introducing next-generation energy sources such as hydrogen are not challenges that can be resolved solely through the corporate efforts of the Company alone, whether in terms of procurement costs or operational aspects. We will continue our dialogue with local governments and other companies as a social mission and will continue conducting demonstration experiments and other activities aimed at decarbonization. The Sustainability Committee was established under my direct oversight as Representative and Chief Executive Officer, and we will continue to engage in deep discussions on various challenges to find the optimal solutions.

Nittobo Is Making Great Strides to the Future, and We Hope You will Join Us

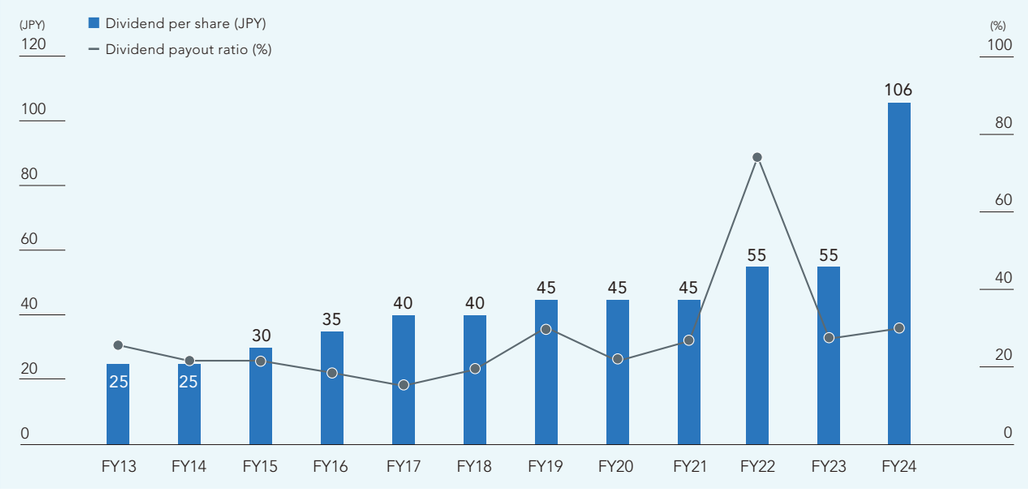
Since 2023, demand for Special Glass has increased, and I now feel that the opportunity has arrived to make the Nittobo Group even larger.

Until now, our creed has been to pursue profit over expanding the scale of the Company. While this policy will remain fundamentally unchanged, we must first build a robust revenue base and firmly seize the opportunity that lies before us. We must work together as one company to seize this opportunity. This means securing funding, generating profits, managing cash flows, effectively communicating the appeal of Nittobo, and establishing structures to attract and retain top human resources.

Together, we must ponder anew what kind of corporate entity we want the Nittobo Group to become, and define the vision we should pursue. The ideal scenario is steadily securing profits from our existing businesses and using that foundation to grow new businesses.

The Company holds even greater potential than when we formulated Big VISION 2030. I hope that each and every employee can feel the momentum of the growing Company. I ask for support from not only our employees but also from all stakeholders, as we look forward to a future in which the Nittobo Group will make great strides, and we hope you will join us for the ride.

Shareholder returns





History of Value Creation

Since its founding in 1923, the Nittobo Group has created many pioneering materials. We were the first in the world to produce glass fiber on an industrial scale (in 1938), and we have continued to grow by taking on the challenge of creating new technologies based on our pioneering spirit. We will continue to anticipate the changing times, help to realize a sustainable society while embracing change as opportunity, and commit to the advancement of industry and society.

Products and services that contribute to the environment and energy

Products and services that contribute to the digital society

Products and services that contribute to health, security and safety

1920-

1923

Nitto Boseki Co., Ltd. established

Fukushima Boseki Co., Ltd. (formerly Fukushima Seiren Seishi) merged with Katakura Seishi Iwashiro Bosekisho Co., Ltd. (formerly Koriyama Kenshi Boseki) to form Nitto Boseki Co., Ltd.
The company name begins with "Nitto," a synonym for Japan, and represents the founder's resolution: "With a vigorous spirit, starting each day anew as a spinning company in the east where the sun rises."

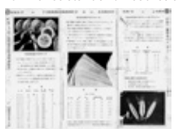


Koriyama Factory (circa 1923–1926)

1938

Succeeded in industrial-scale production of glass fiber (first in the world)
Succeeded in industrial-scale production of rock wool (first in Japan)

We pioneered the industrial-scale production of glass fiber and rock wool, thereby meeting demand for fire resistant and heat resistant materials.


Sales leaflet for glass fiber


Advertising poster (1940)

1949

Started glass wool production

The first in Japan, we started glass wool production using the "spraying method."


Koriyama Factory of Paramount Glass Manufacturing Co., Ltd. (1952)

1960-

1960

Launched Nittobo Dishcloth™

We started selling Nittobo Dishcloth™, which has been a successful product over many years. Nittobo Dishcloth™ won the Good Design Long Life Design Award in 2015.



1964

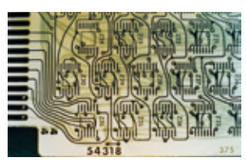
Glass fiber adopted for use in the nose cone for the Tokaido Shinkansen

During the 1960s era of rapid economic growth in Japan, we developed fiber-reinforced polymer (FRP) products, which came to be used in the nose cone for the Tokaido Shinkansen.

1969

Launched glass cloth for printed wiring boards

We started production of glass cloth for printed wiring boards to meet increasing demand in response to advances in computers and integrated circuit (IC) technologies.



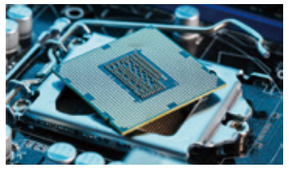
An early printed wiring board

1980-

1984

Launched Special Glass T-glass

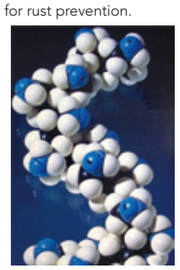
Initial applications for T-glass, characterized by its high strength and low thermal expansion, focused on composite materials. For example, it was used in the Shinkai 6500 manned submersible. Today, T-glass is used in semiconductor package substrates that enable high-speed processing and high reliability in servers and smartphones.


Semiconductor package substrate

1984

Successfully synthesized and launched functional polymer PAA™ (first in the world)

We were the first in the world to develop an industrial-scale production method for polyallylamine (PAA™), which we have patented in the U.S. and many other countries. PAA™ is a functional polymer used in various processing operations such as dyeing, textile treatment, paper manufacturing, and metal surface treatment, as well as for rust prevention.



1987

Established NITTOBO MEDICAL CO., LTD.

After the 1986 acquisition of International Immunology Corporation (IIC) in the U.S., the predecessor of Nittobo America Inc., we established NITTOBO MEDICAL CO., LTD.


IIC's goat farm and factory

1988

Established Nittobo Norplex Oak Co., Ltd.
(predecessor of NITTOBO ASIA Glass Fiber Co., Ltd.)

We established Nittobo Norplex Oak Co., Ltd., in Taiwan as a joint-venture manufacturer of glass fiber yarn.

1998

Launched Special Glass NE-glass

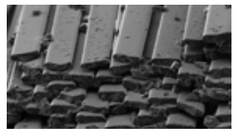
NE-glass is used in switches and routers in supercomputers and data centers that process large volumes of data at high speed, and in antenna-in-package modules for 5G smartphones.

2000-

2006

Launched Flat Fiber™

Launched chopped strands made of modified cross section fiber developed using proprietary technology. Flat Fiber reduces curling and warping in molded products and is used in smartphone cases and automobile parts.



2018

Launched IgG4 measuring reagent kit for use with automated analyzers

Launched IgG4 measuring reagent kit for detecting IgG4-related disease (designated an intractable disease in Japan) with automated analyzers.



2020

Established NI-CoLabo in the Research & Development Headquarters (current R&D Building)

Established the NI-CoLabo facility at the Fukuyama Enterprise Center in Koriyama City, Fukushima, to promote timely delivery of high-value-added, highly original products and services that harness the Group's characteristics and strengths, and to pursue forward-looking research and technological development.



2021

Established Special Glass factory at NITTOBO ASIA Glass Fiber., Ltd. (Taiwan)

Established Special Glass Yarn factory at NITTOBO ASIA Glass Fiber Co., Ltd. (Taiwan) Established an integrated production system from Special Glass yarn to cloth at the factory on the main island of Taiwan.



2023

100th anniversary

since1923
Nittobo

2024

Started full-scale development of Special Glass NER-glass

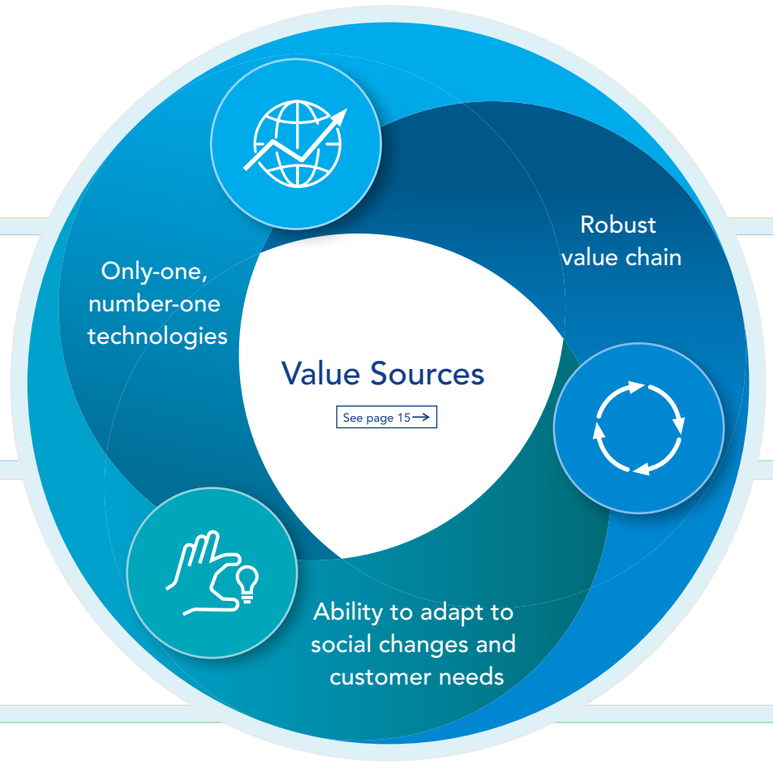
Started mass production of NER-glass, a second-generation low-dielectric glass that significantly improves the performance of data center communications equipment supporting a high-speed, large-capacity communications society, and AI-related equipment.

Value Creation Process

Corporate Philosophy

The Nittobo Group will enhance its value to society as a corporate group by helping to create healthy and more comfortable lifestyles, and in so doing continue contributing to the realization of a more prosperous world.

Social Issues	Material Issues	Business Activities	Outputs	Outcomes
Intensification of global warming and environmental issues • Transition to a recycling-oriented society • More rapid legal and regulatory change			Lightweight, energy-efficient automobiles and aircraft ● Chopped strands ● Glass fiber cloths Energy-efficient housing, recycling ● Glass wool insulation Wind power ● Chopped strands Wastewater treatment ● Specialty chemicals	Supplying products that reduce environmental burden
Transition to an ultra-smart society • Higher speed telecommunications (5G/6G) • Evolution of AI technologies and start of the IoT era			High-speed, large-volume data communications Network infrastructure connecting everything ● Special Glass cloths Smaller, thinner electronic devices ● Ultrafine glass yarns, ultra-thin glass cloths Infrastructure repair and reinforcement ● Non-woven fabric, glass fiber cloths Sustainable auxiliary materials ● Interlinings, inner materials	Creating a livable, convenient and safe advanced digital society
Societal structure issues • Aging society with a declining birthrate (in Japan) • Population and other issues			Supporting the shift from treatment to prevention ● In vitro diagnostics reagents Fashion and functionality ● Deodorizing and antibacterial interlinings ● Innovative Fabric™	Realizing a healthy, safe and secure society
• Increased geopolitical risk • Industry reorganization, entry from other industries • More sophisticated and complex social demands • Increasing pace of innovation			Harmonious Coexistence with Society Safety initiatives Active Roles for Diverse Talent	Contributing to the realization of a sustainable society



Continuously updating corporate governance

Medium-Term Management Plan
See page 19

● Electronic Materials Business Division
● Medical Business Division
● Composite Materials Business Division
● Materials Solution Business Division
● Insulation Materials Business Division

Inputs

See page 14

Manufacturing capital | Intellectual capital | Human capital | Social and relationship capital | Natural capital | Financial capital

SUSTAINABLE DEVELOPMENT GOALS

Nittobo Group's Vision for 2030
Big VISION 2030

Global No.1 Niche Company

Value Creation Drivers and Capital

The Nittobo Group's strengths, established over its history of more than 100 years, are the drivers of value creation that meets contemporary needs, helping the Company to stay abreast of environmental changes and paving the way to continued growth. As we continue to grow, we will create greater value by leveraging these strengths to resolve social issues.

Forms of Capital at the Nittobo Group (FY2024)

Manufacturing capital

A safe and secure production system for high-value-added products

Characteristic

Through bold capital investment we have established a supply system capable of meeting growing demand for Special Glass and in vitro diagnostic reagents.

Initiative

We are establishing new production systems for fields where market growth can be expected. We will continue to actively invest for growth to strengthen our supply capacity so that we can respond flexibly to sudden increases in demand.

Capital expenditure
JPY 13.6 billion

Intellectual capital

Technical capabilities that create Global No. 1 Niche Businesses

Characteristic

Inspired by our motto, "Let's turn anything into fibers," and ever eager for a challenge, our high-level R&D consistently leads to industry first technologies and products, both in Japan and in the world.

Initiative

We are developing the knowledge we have accumulated so far while also engaging in seeking out new knowledge. We are pursuing the establishment of an advanced and flexible R&D structure centered on open innovation and digital innovation.

R&D expenses
JPY 3.0 billion

Patents held
737

Human capital

Human resources that innovate to meet new needs

Characteristic

We are committed to fostering a culture where the skills and approaches of individual employees are valued and that inspires them to freely share and discuss their ideas, ultimately achieving greater results through teamwork.

Initiative

We are working to enhance employee engagement, promote digital transformation, develop global human resources, and create an environment conducive to an appropriate work-life balance.

Total employees
2,745

Social and relationship capital

Strong relationships rooted in the front lines

Characteristic

The Nittobo Group was founded in Fukushima Prefecture. In addition to maintaining multiple factories and research and development facilities in the Prefecture, we conduct production activities throughout Japan, providing products and services while placing importance on relationships with regional communities at each of our production sites.

Initiative

In addition to maintaining and developing our domestic production system, we are expanding our production system to better meet customers' needs globally, with a glass fiber facility in Taiwan and an in vitro diagnostic reagents facility in the U.S. and strengthening our supply system in overseas markets.

- Excellent relationships with customers and suppliers
- Strong sense of stakeholder trust

Natural capital

Reducing environmental impact to help create a sustainable society

Characteristic

Our production processes have an industrial characteristic of requiring substantial amounts of fuel and electricity.

Initiative

In keeping with our responsibility as a manufacturer, we strive to reduce CO_2 emissions as a means of reducing environmental impact, such as by using renewable energy. We also leverage our products and technologies to help resolve environmental issues throughout society.

Energy input
5.41 million GJ

Financial capital

Finance management underpinning our growth strategy and healthy financial position

Characteristic

While making necessary investments for growth and next-generation product development, we maintain a healthy financial position, thereby ensuring our growth is sustainable.

Initiative

In order to steadily pursue our growth strategy, we are focusing on securing financial soundness while investing funds with our sights set on the future, with the aim of continuously enhancing our corporate value.

Net interest-bearing debt
JPY 24.2 billion

Net assets
JPY 135.8 billion



Value Sources

Only-one, number-one technologies

Our consistent quality is supported by our ability to develop high-performance products, and the wealth of manufacturing technology and expertise we have amassed over many years.

- Ability to develop Special Glass made from proprietary compositions
- Development and production technology of goat antisera and in vitro diagnostic reagents for plasma proteins
- First in the world to industrially produce modified cross section fiber
- Functional polymers developed through proprietary technologies
- Glass wool with high insulation performance from proprietary technologies

Ability to adapt to social changes and customer needs

We will establish a unique position throughout our product development and deepen our relationships with global clients.

We are committed to providing products and services that contribute in the areas of Environment & Energy, Digital Society, and Health, Security & Safety. Harnessing the potential of all ideas to achieve pinpoint responses to needs, we will deploy strategies based on our technical strengths in new fields and develop products that are indispensable to society.

Robust value chain

Through an integrated production system incorporating upstream and downstream processes, we maintain a stable supply of high-performance, high-quality products.

Glass fiber

Composition research → Spinning processes → Weaving processes → Interface treatment → Product sales

In vitro diagnostic reagents

Antigens/Antibodies → Antisera → Reagent development and production → Reagent sales

Nittobo Group's Material Issues

In order for the Nittobo Group to enhance its social value and continue contributing to the realization of a more prosperous world, we have identified material issues as important management tasks that will contribute to resolving social issues.

We have defined seven material issues by applying the perspectives of strategic CSR, which leverages the strengths of the Nittobo Group, to contribute through business activities, and of basic CSR, which focuses on ESG matters to build a foundation for sustainable growth.

Materiality Identification Process

1 Issue Selection
- Discussed the Nittobo Group's purpose and value creation

2 Analysis and Plan Creation
- Business divisions analyzed their respective environments and formulated a vision for the future
- Held multiple intensive discussions

3 Management Decision-Making
- Discussed in depth by the Board of Directors with external directors' input and provided feedback to the business divisions

4 Setting and Development of Material Issues
- Defined seven material issues
- Developed specific initiatives to realize our vision

Social Issues	Nittobo Group's Material Issues	Risks	Opportunities	Key Measures	Relevant SDGs
Intensification of global warming and environmental issues • Transition to a recycling-oriented society • More rapid legal and regulatory change	Provide products and services that contribute to the environment and energy	• Increase in environmental costs • Restrictions on business due to environmental regulations • Exclusion of products and companies that are not environmentally responsible	• Increasing demand for sustainable products • Expanding demand for composite materials to meet the needs of lightweight automobiles and aircraft • Increasing demand for insulation materials due to stricter energy conservation regulations	• Develop products that contribute to the environment	
	Adapt to climate change and a resource-recycling society			• Reduce CO_2 emissions • Reduce glass waste	
Transition to an ultra-smart society • Higher speed telecommunications (5G/6G) • Evolution of AI technologies and start of the IoT era	Provide products and services that contribute to the digital society	• Decline in technological superiority • Replacement by other materials • Speed of technological development in the electronic materials industry	• Increasing demand for devices that require high-speed, high-capacity processing • Utilization of the production system in Japan and in Taiwan, where electronic materials are concentrated • Opportunities to expand sales and develop next-generation products through first-mover advantage	• Grow profits through Special Glass • Strengthen development capabilities for new products • Strengthen solution capabilities to increase value for customers	
Societal structure issues • Aging society with a declining birthrate (in Japan) • Population and other issues	Provide products and services that contribute to health, security and safety	• Shrinking domestic market due to declining birthrate • Business continuity risks specific to biologically derived materials • Replacement by other diagnostic reagents	• Global shift from treatment to prevention due to increased awareness • Expanding demand for diagnostic reagents globally due to population growth and rising income levels • Utilization of three raw material production bases in the U.S. • Utilization of the value chain from upstream (antigens, antibodies, raw materials) to downstream (product manufacturing and sales)	• Expand sales channels in the field of in vitro diagnostic reagents • Accelerate research and development, and promote new product development • Strengthen solution capabilities to increase value for customers	
• Increased geopolitical risk • Intensifying competition due to industry reorganization, entry from other industries, etc. • More sophisticated and complex social demands • Increasing pace of innovation	Realize continuous, sustainable growth for the Company				
	Strengthen management base	• Diversification of factors causing changes in the socioeconomic environment • Price fluctuations of fuels, raw materials and other commodities • Impact of geopolitics and human rights issues, etc. on the supply chain	• Selection and concentration • Capture growth opportunities through innovative technology development	• Lean management that is not substantially affected by economic fluctuations • Optimize business portfolio and review unprofitable businesses • Transform technology development and production technology through the introduction of IT and digital transformation • Maintain sustainability in procurement and the supply chain	
	Secure human resources and create job satisfaction	• Decline in workforce due to low birthrate and aging population • Intensifying competition to recruit young and specialized human resources	• Increasing diversity • Increasing mobility in the labor market • Innovation expected through human resource development	• Foster human resources that generate innovation • Implement diversity and inclusion measures • Conduct workstyle reform and business reform • Improve employee engagement • Strengthen external recruitment • Utilize senior human resources	
	Continuously update corporate governance	• Expanding and refining CSR scope • Compliance and security in a remote environment	• Permeation of corporate philosophy • Strengthening the corporate foundation by gaining trust from society	• Continuously update corporate governance • Thoroughly implement corporate ethics and compliance	

Big VISION 2030

Big VISION 2030 is the Nittobo Group's long-term vision for 2030. It outlines our initiatives for realizing a sustainable society based on our contributions in Environment & Energy, Digital Society, and Health, Security & Safety—areas where the Group can contribute to resolving social issues as a Global No. 1 Niche Company. We aspire to be a company that gains the trust of all stakeholders in the pursuit of client satisfaction.

Nittobo Group's Vision for 2030

Big VISION 2030

As a corporate group that continues to create

Global No. 1 Niche Businesses

we aspire to contribute in the areas of

Environment & Energy | Digital Society | Health, Security & Safety

with the goal of building a more sustainable society.

We aspire to be a company that gains the trust of all stakeholders in the pursuit of client satisfaction.

Provide products and services that contribute to the environment and energy

Provide products and services that contribute to the digital society

Provide products and services that contribute to health, security and safety

Nittobo Group as Global Niche No. 1

Pinpoint response to needs	Development of products indispensable to society
Extract the full potential of ideas to uncover latent demand	Leverage our technological strengths to also take on the challenge of new fields

Establish a unique position throughout our product development and deepen our relationships with global clients

Medium-Term Management Plan (FY2024–2027)

Key Points of the New Medium-Term Management Plan

The Medium-Term Management Plan centers on two key points.

The first point is steadily realizing returns on investments (made during the previous medium-term management plan period) in the glass fiber for electronic materials field and the medical field, as we work to realize Big VISION 2030. We will also continue to actively invest for growth to ensure that our supply structure is able to respond to rapidly expanding new markets.

The second point is building new pillars of revenue for the next 100 years beyond 2030. We will restructure our previous three business divisions into five business divisions; expand into new fields to follow Special Glass and the medical field; integrate product development, manufacturing, and sales; and strengthen activities that consider the customer's perspective.

POINT 1

Steadily harvesting the fruits of investment and promoting existing growth strategies to realize Big VISION 2030

We will steadily realize returns on investments in Special Glass and the medical field made during the previous medium-term management plan, and continue to actively invest for growth to enable us to respond to rapidly expanding new markets

POINT 2

Building new pillars of revenue for the next 100 years beyond 2030

- Building additional pillars of revenue to follow Special Glass and medical
- Unified management of product development, manufacturing and sales, strengthening of customer perspective

▶ Launch of new structure

Company-wide Quantitative Targets and Progress

	Results for FY2024	Target
Net sales	JPY 109 billion	JPY 135 billion
Operating profit	JPY 16.4 billion	JPY 20 billion
EBITDA	JPY 24.4 billion	JPY 32 billion
ROE	10.4%	8% or higher (a level consistently exceeding cost of capital)
Capital investment (4-year total)	JPY 13.6 billion	Approx. JPY 80 billion
R&D expenses (4-year total)	JPY 3 billion	Approx. JPY 15 billion
Net D/E ratio	0.19 times	0.4 times or lower
Equity ratio	58.1%	55% or higher
Shareholder return policy	Our basic policy is to pay a dividend of not less than JPY 55 per share, targeting a dividend payout ratio of 30% of steady earnings	
Establishing incentives for employees	An additional third bonus will be paid when operating profit targets are achieved on a single-year basis. At JPY 12.5 billion, an additional 0.5 months' worth of the monthly base rate will be paid, increasing by 0.5 months for every additional JPY 2.5 billion thereafter.	

In FY2024, sales of high-value-added products was strong, particularly in the Electronic Materials Business. As a result, consolidated net sales reached JPY 109 billion (a 16.9% YoY increase), consolidated operating profit reached JPY 16.4 billion (a 96.1% YoY increase), and EBITDA reached JPY 24.4 billion. We also enhanced the research and development of high-performance glass fiber for electronic materials, for which demand is expected to grow further, and new products.

Sustainability Initiatives

Human Resources

To promote sustainable growth, we are promoting efforts to secure human resources and to create job satisfaction

FY2030 Targets
- Secure human resources: 20% increase globally by FY2027
- Create job satisfaction: Improve employee engagement

Regardless of gender, age, nationality, or other factors, we attract and acquire diverse human resources while implementing various initiatives to ensure these individuals have a sense of job satisfaction and are able to work wholeheartedly.

Environment

Responsible manufacturing initiatives to help create a sustainable society

FY2030 Targets
- Reduce CO_2 emissions by 30% (compared to FY2013)
- Achieve net-zero waste glass
- Develop products that contribute to the environment

In response to the expected increase in emissions due to expanded production capacity, we are implementing measures to reduce CO_2, such as introducing a cogeneration system (CGS) and utilizing renewable energy. In Environmental Contribution Product Development, we are promoting initiatives such as creating new products using recycled glass.

Corporate Governance

We are constantly updating our structure to enhance management transparency and further strengthen our organizational framework

FY2030 Targets
- Maximize corporate value and achieve sustainable growth
- Strengthen our corporate governance structure

To invigorate deliberations at board meetings, we provide external directors with timely and appropriate explanations and opportunities for exchange of views, creating a forum for them to deepen their fundamental understanding of the background of proposals.

NEWS



New factory building render

Regarding Expansion of Glass Cloth Production Capacity

In August 2025, Nittobo decided to expand its glass fiber cloth production facilities at the Fukushima Enterprise Center. Production is scheduled to begin in the fourth quarter of FY2026.

With the rapid evolution and widespread adoption of generative AI technology in recent years, AI servers are becoming more powerful and their market size is expanding. Since 2024, advanced packaging technologies enabling high-density connections of multiple chips for AI server semiconductors have begun to be widely adopted. While these technologies make sophisticated semiconductor packages possible, they also require larger semiconductor package substrates and present challenges such as thermal expansion and warping. To respond to these issues, the Company's T-glass fiber cloth has been highly evaluated as an effective solution, and demand continues to grow. This trend is expected to continue, and we have decided to expand our production facilities to meet this increasing demand.

Since FY2017, we have been promoting a high-value-added strategy and working to enhance our Special Glass production capacity. Under the Medium-Term Management Plan (FY2024–2027), we plan to invest JPY 80 billion in facilities over four years. Looking ahead, we will continue making necessary investments while carefully monitoring market trends, aiming to establish ourselves as the "Global No. 1 Niche Company" in electronic materials.



Aerial rendering of the factory building



Groundbreaking ceremony held on Wednesday, September 17

Message from the Division General Manager of the Corporate Management Division



We will pursue results with a strong awareness of the future, balancing short-, medium-, and long-term management.

Yasushi Okahisa

Managing Executive Officer, Division General Manager of the Corporate Management Division

In Addition to Creating Synergies Between Businesses, Our Mission Is to Balance Centrifugal and Centripetal Forces

In the Company's business, it can take on the order of ten years for a single product to go from development to contributing to profits.

The Corporate Management Division strives to maintain and reinforce the Company's fundamental ability to persist through long-term projects. At the same time, we continuously evaluate whether there are changes in long-term direction. If changes are anticipated, we carefully consider what countermeasures are necessary and plan the necessary course corrections.

Following the reorganization of our structure into five business divisions, a key mission of this Division is to support the operations of each business division while also ensuring cross-functional coordination. Our agility has been enhanced and our centrifugal force increased by reorganizing the organization from a customer perspective. At the same time, centripetal force as the Group is also necessary. We recognize that our role is to strike a proper balance between the two. We consciously use our two functions of planning and management, either stepping on the accelerator or applying the brakes, to connect and unite the entire Group.

Under the current Medium-Term Management Plan, we are working to build the third and fourth pillars, following on the heels of Special Glass and the medical field. This is a major challenge. The Company will maintain and strengthen a system that allows us to fully leverage the deep knowledge and extensive experience possessed by its employees to develop new businesses from a long-term perspective.

Expanding the Unique Strengths of Each Business Division and Building New Pillars of Profit

In FY2024, the first year of our current Medium-Term Management Plan, we achieved profits above our initial projections and got off to a strong start. This is the result of years of research and development, combined with the significant investments made over the past several years. Our business structure involves initiatives that take time to bear fruit, so measures implemented during the four-year period of the current Medium-Term Management Plan will contribute to results leading up to 2030.

In our Medium-Term Management Plan, we set KPIs as commitments to our stakeholders, and we achieved the FY2024 targets. While continuing to prioritize achieving KPIs, we also need to take necessary risks at the appropriate times. This may result in a temporary increase in leverage or impact ROE for a single fiscal year, but we recognize the importance of being fully prepared for such situations.

A year has passed since the transition to the new system, and each business division has gradually developed its own unique character, to the point where they can be seen as their own mini-companies. We have significantly delegated authority to business divisions, which has increased the speed of our operations. However, we will continue to explore ways to further enhance the agility of our business.

We will continue to work tirelessly and diligently on the daily operations of our business, aiming to achieve significant results. Please look forward to what we will accomplish next.

Message from the Division General Manager of the Technology Development Division



We will foster a corporate culture and environment where employees can take on challenges without fear of failure, promoting the pursuit of innovation

Koichi Nakamura
Executive Officer
Division General Manager of the Technology Development Division

One Year Since the Launch of the Technology Development Division: Our Current Position and Signs of Further Change

It has been one year since the Technology Development Division was reorganized from the Research & Development Headquarters in April 2024 in line with an organizational structure change. This was a major reorganization that restructured product development functions directly under each business division. Young engineers have been able to see the results of their work more clearly thanks to more agile work practices and by becoming closer to customers, which in turn has boosted their motivation.

There were concerns about whether the Technology Development Division, which specializes in basic research and production technology, could continue promoting information sharing and collaboration with product development divisions. Although the organizational structure has changed, communication and horizontal collaboration have been revitalized through weekly and monthly technical liaison meetings, which now include representatives from the Materials Solution Business Division and Insulation Materials Business Division. In addition, mechanisms have been established to promote cross-divisional themes, beginning with technical exchanges across different industries. Collaboration centered around the Production Technology Department and the R&D Planning Group is also advancing. An example of ambidextrous development in the field of electronic materials is how we always keep in mind the milestones of the next-generation information and communication technology roadmap and carry out everything seamlessly, from basic research and prototyping to mass production. While ensuring that we preserve the good aspects of the past, we have begun to see activities that go beyond the traditional framework. During this first year since the transition to the new structure, we have gotten a good start on integrating development, manufacturing, and sales. We have become able to promote a proactive IP (intellectual property) strategy utilizing patent maps, such as applying for patents for peripheral technologies once the direction of product development has been determined to a certain extent.

Redefining Core Technologies and Accelerating Innovation and Cross-Sector Exchanges in Adjacent Fields

A characteristic of Nittobo's technological development is that it judges based on the two axes of technological risk and market risk. It focuses on innovation in adjacent fields that are extensions of existing customers and existing technologies, rather than completely unexplored areas. This area was selected because it is where Nittobo's strengths and scale allow us to respond to market needs.

We have set the goal of creating one or two new technologies in adjacent fields by 2030. To achieve this, we have redefined our core technologies and identified multiple markets based on their similarities and potential applications. We have also established a system to develop strategic plans based on market size and technology portfolio analysis, with the R&D Planning Group playing a central role. Investigations are currently underway into the potential of various fields, including contributions to solving social issues (energy issues/SDGs) and materials related to the space business sector, which we have not covered before.

When it comes to fostering innovation, external exchanges provide the greatest stimulus. The cross-sector exchanges that began in 2023 initially focused on joint research themes in the field of electronic materials but have since expanded to include research on innovative materials such as healthcare and nanomaterials.

At technological exchange meetings for junior employees, junior engineers of the same age group from different companies share issues and success stories, and by learning from each other, improve their skills and build networks. This serves as an opportunity for them to take an objective reassessment of their own company's situation and helps them maintain a high level of motivation.

Promoting Challenging Initiatives Toward True Process Innovation

While the glass fiber manufacturing method, a core process, efficiently and consistently produces high-quality products, it faces various issues, such as the need to improve energy conservation, product life, and workloads. We sincerely addressed those issues, set unprecedentedly high goals, and began technological development and initiatives with our factories to achieve them. In addition to our high-value-added strategy based on high-performance products, we aim to maintain high profitability by engaging in unprecedented cost management and talent development. This involves reviewing all energy, raw material, and logistics costs from a technical perspective and applying backcasting based on our targets.

The Technology Development Division is responsible for basic research and production technologies, and also for DX and environmental technologies. Organically connecting and expanding these was a major challenge, but the collaboration is beginning to take shape, and we believe it will bear fruit in the near future. The number of DX literacy education graduates has increased, providing them with not only knowledge of digital technology in general but also a base for properly evaluating the information they obtain and flexibly adapting to new technologies and digital environments. With the widespread awareness of the potential of digital technology and a heightened awareness of the risk of labor shortages in the future, there has been an increase in initiatives to save labor and automate processes, such as RPA* and robotics.

We have established a new R&D portal on our intranet so that various activities and successful case studies can be shared across the organization as common values among all employees. Our junior engineers are talented and full of potential, and we will create environments where they can freely come up with ideas and propose plans.

* RPA (Robotic Process Automation): An initiative to automate routine tasks previously performed by humans, using cognitive technologies such as rule engines, AI, and machine learning.

Continuing To Be a Nittobo That Boldly Takes on Challenges Without Fear of Failure

With our corporate philosophy of "helping to create healthy and more comfortable lifestyles," the Company aims to be a company that is loved by everyone and needed by society for the next 100 years as the Best Partner for society. We are focusing on two areas in particular: digital infrastructure and energy infrastructure, with a view to achieving carbon neutrality and building a circular economy.

To achieve these goals, we will return to the spirit of our founding days, which was to relentlessly challenge ourselves without fear of failure, and we will continue to explore new frontiers. A corporate culture and environment that encourages taking on challenges is the source of everything, and it is my mission as Division Manager of the Technology Development Division to foster such a culture and environment.

Opening Up a New Future Based on Three Themes with the Objective of Realizing the Long-Term Management Strategy

(1) Ambidextrous Development

- Deepen knowledge by innovating the production and technology foundation:
  - Strengthen and implement the technology foundation that will be required in the future
- Explore knowledge creation by utilizing and redefining core technology:
  - Sow seeds for/open up new markets

(2) Open Innovation

- Co-create with external partners:
  - Respond promptly to increasingly complex social issues
- Develop human resources and create new themes through exchanges with different industries:
  - Free innovation activities

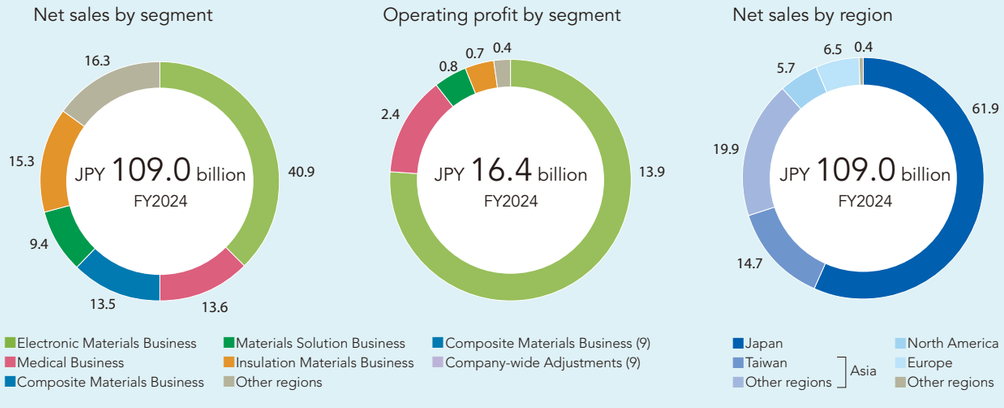
(3) Digital Innovation

- Tackle digitalization by taking a bird's-eye view of various business processes including those in technology development, production sites, and the back office, and use current projects utilizing MI* and AI (i.e., implementing AI) to open the way to process innovation (stage before shifting to CPS)

*** MI:** Materials informatics. A field that aims to enhance the efficiency of material development by applying informatics methodologies, such as statistical analysis.

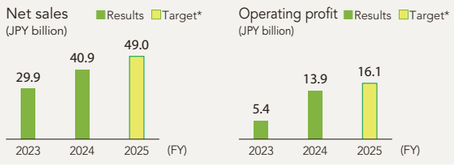
Business Overview

Net Sales and Operating Profit



Electronic Materials Business Division

Develops, manufactures, and sells glass yarn and glass fiber cloth for electronic material applications

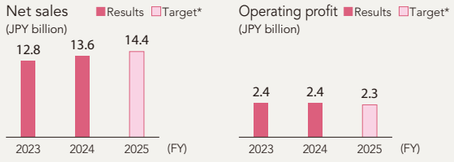


Overview of FY2024

Due to the continued strong demand for AI servers, sales of special glass remained strong and contributed to profits.

Medical Business Division

A medical business centered on in vitro diagnostic reagents

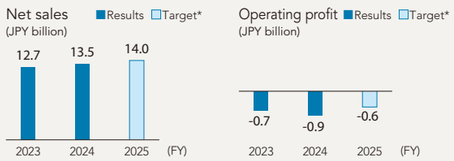


Overview of FY2024

Although sales of in vitro diagnostic products remained strong, they were affected by global inflation.

Composite Materials Business Division

Develops, manufactures, and sells composite materials, mainly glass fiber for plastic reinforcement material applications

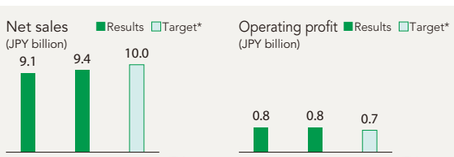


Overview of FY2024

Although sales exceeded the same period last year, they were affected by cost increases associated with the regular maintenance of production facilities.

Materials Solution Business Division

A glass fiber for industrial materials business, chemical business, and textile business

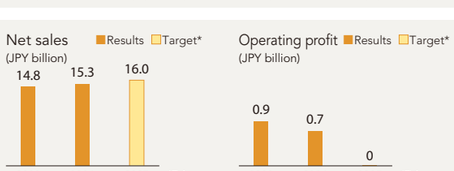


Overview of FY2024

Although affected by cost increases mainly for raw materials, sales exceeded the previous period, partly due to price increases.

Insulation Materials Business Division

Develops, manufactures, and sells glass short fibers for use in equipment materials, mainly insulation



Overview of FY2024

Although sales of high-performance insulation remained strong, they were affected by factors such as cost increases.

* Target figures are accurate as of August 1, 2025.

Electronic Materials Business

Vision

Promote the shift to next-generation products and aim for further improvements in net sales and profitability

Issues and Initiatives
1. Harvesting fruits of investments executed during the previous medium-term management plan
2. Strengthen measures to address growing demand for Special Glass
3. Strengthen marketing activities and research and development of next-generation Special Glass

In the Electronic Materials Business Division, we manufacture and sell glass fiber yarn and glass fiber cloth, and have been expanding our business centering on the Special Glass, developed through our proprietary technologies.
Our Special Glass is used as an insulating material for high-performance printed wiring boards used in high-speed, high-capacity applications that require materials with properties such as a low dielectric constant, a low dielectric dissipation factor, and low thermal expansion. Moreover, its quality has gained widespread global recognition. In order to meet strong demand for Special Glass, we are working to improve productivity and making capital investments to increase production capacity.

Strengths	• Manufacturing technology that enables stable production of Special Glass • Product supply capacity from ongoing capital investment • Unique product development based on our development roadmap
Risks	• Large fluctuations in demand due to the impact of the semiconductor cycle • Possibility of replacement by alternate materials as the result of technological innovation
Opportunities	• Expanding demand for our Special Glass, which can support infrastructure for high-speed, high-capacity communications • Increasing demand for our products that can be used in the AI-related market due to market expansion of AI-related businesses and other factors

Main Products

Yarn



Nittobo manufactures and sells various types of single yarn for printed wiring boards and industrial materials. Its unique product lineup, which includes several types of glass fiber yarn, and a range of linear densities (tex), yarn counts, and filament diameters, meets customers' needs.

Glass fiber cloth



Our specially treated glass fiber cloth, woven from glass fiber yarn, is used as an electrical insulator for isolating circuits and serves as a reinforcement material that contributes to substrate strength.

Supply Chain

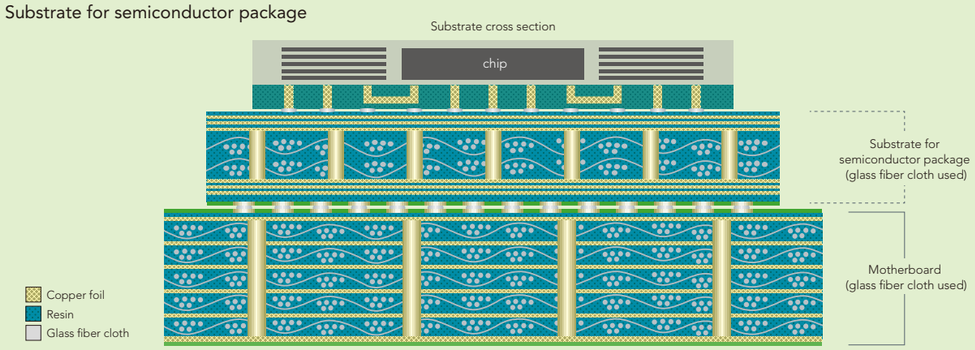
■: Nittobo Group

Glass fiber yarn → Glass fiber cloth / Glass fiber cloth manufacturers → Copper-clad laminate (CCL) manufacturers → Printed wiring board manufacturers → Semiconductor manufacturers → Electronic devices

Typical Application for Glass Fiber Cloth

Glass cloth provides excellent insulation, heat resistance, and dimensional stability, and is used as a substrate for printed wiring boards.

Substrate for semiconductor package



Medical Business

Vision

Become world No. 1 in the field of plasma proteins and bones

Issues and Initiatives
1. Boost global expansion
2. Accelerate research and development, and promote new product development
3. Strengthen manufacturing and sales at Nittobo America Inc.
4. Reinforce foundation

The Medical Business Division's operations are centered on in vitro diagnostic reagents. They are used in health checkups and at hospitals to test blood and urine and determine state of health. The Group's global value chain, which integrates everything from raw materials production in the U.S. to completed products, enables us to offer a stable supply of high-quality in vitro diagnostic reagents and has earned us the top domestic market share for many items, including inflammation markers. In particular, our in vitro diagnostic reagents for immunological plasma proteins have helped us establish a reputation as specialists in immunology among healthcare professionals throughout Japan and worldwide.

Strengths	• A global value chain, from raw material production to marketing of reagents • Solid presence in the market for in vitro diagnostic reagents for immunological plasma proteins
Risks	• Unforeseeable issues may hinder our ability to maintain stable quality • Impact of global economic trends, trade and tax policies, and geopolitical factors
Opportunities	• Increased demand from overseas markets for in vitro diagnostic reagents • Earning suppliers' trust creates opportunities for solution proposals

Main Products

In vitro diagnostic reagents

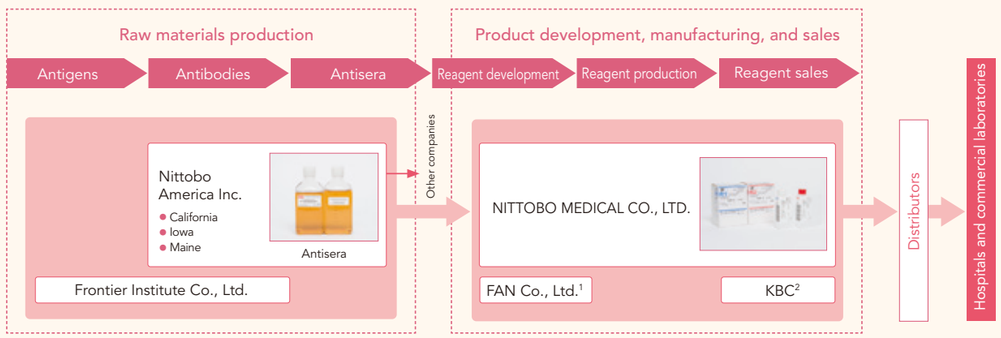


Our in vitro diagnostic reagents are used in health checkups and at hospitals to test blood and urine and determine state of health. Establishing an integrated production system from raw materials to completed product has enabled us to offer a stable supply of high-quality in vitro diagnostic reagents.

Main Products

Main test target	Test item
Inflammation	CRP
	IgG/IgA/IgM
	C3/C4
Diabetic renal disease	Urinary albumin
Glomerular filtration rate	Cystatin C
Renal tubular disorder	β2 microglobulin
Nutritional status	PreALB/RBP/Tf
Allergies	IgE
Rheumatism	MMP-3/RF
Streptococcal infection	ASO
Iron metabolism	Ferritin/Iron/UIBC
Osteoporosis	TRACP-5b
IgG4-related disease	IgG4

Global Value Chain



1. Joint venture with Fujikura Kasei Co., Ltd. 2. Kamiya Biomedical Company, LLC

Composite Materials Business

Vision

Without being bound by fixed ideas, we will achieve the following targets by taking on the challenge of change
- Higher productivity; greater cost competitiveness
- Develop products and processes that contribute more widely to society
- Build closer relationships of trust with customers

Issues and Initiatives
1. Improve productivity and strengthen cost competitiveness
2. Develop products and explore demand from a customer perspective
3. Develop environmentally friendly products and production processes

Our Composite Materials Business Division supplies glass fiber as a plastic reinforcement material. Our glass fibers offer high shape stability and ease of processing, and are used mainly for automotive components, smartphone-related components (e.g., housings), and residential equipment (e.g., bathtubs).
We offer products in a broad range of categories, including continuous fiber roving, uniform length chopped strands, and modified cross section high-performance glass, which was developed using Nittobo's proprietary technologies. Our products are used to meet a variety of needs related to fiber-reinforced plastics (FRPs) and fiber-reinforced thermoplastics (FRTPs).

Strengths	• Ability to meet customers' needs with an extensive lineup of products based on our proprietary glass-composition technologies
Risks	• Deterioration of productivity and cost-competitiveness due to factors such as intensifying competition • Threat to business performance due to external factors such as rising raw material and fuel prices
Opportunities	• Rise in demand for use of recycled raw materials and environmentally friendly products, driven by increasing environmental awareness

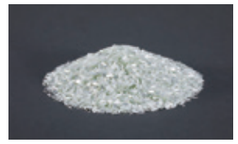
Main Products

Roving



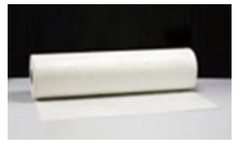
Roving is the simplest form of glass fiber and has a wide range of applications as a fundamental component of composite materials. We offer strands that are processed optimally according to the application and manufactured to specific yarn counts.

Chopped strands



With a focus on molding techniques and affinity with resin, we continue to develop our chopped strands, which can be used for diverse applications. We meet a variety of needs by offering a wide range of treatment formulations, strand types, and cut lengths.

Chopped strand mat



Chopped strand mats increase affinity with various kinds of resin, and thereby enhance the strength and boiling resistance of molded products. Tape mats, meanwhile, which are cut according to tape shape, facilitate corner reinforcement, overlay, adhesion, and other sorts of work.

Cut fiber



Cut fiber consists of powdered glass fiber. Filling resin with the product is expected to produce such effects as dimensional stability, crack resistance, surface modifiability, and improved heat resistance.

FlatFiber

Modified Cross Section Chopped Strands:

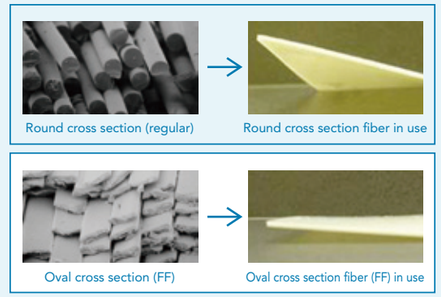
Flat Fiber

Nittobo's proprietary high-performance Flat Fiber (FF) has an oval cross section, and offers high strength and dimensional stability. Its applications include smartphone housings and automotive components.

Characteristics of Flat Fiber

High flowability | High strength | Surface smoothness | Impact resistance | Dimensional stability

Warp Level Comparison for Composite Panels Made with Flat Fiber



Materials Solution Business

Vision

Create new businesses through collaboration and flexible thinking that is not bound by the framework of existing operations

Issues and Initiatives

1. Create new businesses
2. Create an organization that generates synergies
3. Acquire new commercial rights in growth areas

The Materials Solution Business Division incorporates the glass fiber for industrial materials business, the chemical business (including functional polymers), and the textile businesses.
In the Glass Fiber for Industrial Materials Business, we provide products that take advantage of glass fiber's heat resistance, fire resistance, and light weight, such as architectural membranes for large buildings and automotive vibration-damping materials.

In the chemical business, we are engaged in the development and sale of highly original functional polymers and organic-inorganic hybrid materials (SSGs) that are used in various applications to support daily life. In the textile business, we leverage our proprietary adhesive technologies developed through experience in adhesive interlinings to provide functional materials not only for conventional clothing applications but also for use in a broad range of other business domains.

Strengths	• High strength and high heat resistance membranes • World-leading broad-cut membranes • R&D capabilities for new functional polymers leveraging proprietary raw materials and manufacturing expertise • Ability to make functional materials without compromising substrate properties, backed by proprietary adhesive technologies developed in the textiles business
Risks	• Deterioration of productivity and cost-competitiveness due to factors such as intensifying competition • Threat to profitability due to the impact of external factors such as rising raw material and fuel prices
Opportunities	• Responding to the diversification of customers' needs by creating new businesses through collaboration and flexible thinking that is not bound by the framework of existing operations

Main Products

Glass fiber for industrial materials

With excellent heat resistance, fire resistance, and mechanical properties, these products are used widely as architectural membranes for stadiums and film materials for tent warehouses, as well as temporary structures for various kinds of events.

Main Applications



Roll-up blinds



Architectural membranes



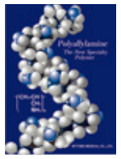
Vibration-damping materials



Coating film waterproofing

Chemicals

The Group engages in the development and sale of proprietary functional polymers such as Poly-allylamine and the Polyamine Series. We research, develop, manufacture and sell proprietary products in categories with high barriers to entry and across a wide range of industry sectors, including toiletries, papermaking, metals, electronic materials and generic drugs.





Textiles (adhesive interlinings)

We also have a large share of the Japanese market for interlinings for high-end ladieswear. Interlinings are an auxiliary material used between the inner and outer fabrics of, for instance, jackets and overcoats, to help them retain and adjust shape.



Textiles (dishcloth)

Created through joint research with Kurashi-no-techo Co., Ltd., Nittobo Dishcloth™ is a long-selling product whose popularity stretches back to its launch more than 60 years ago.



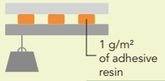
Features of Our Adhesive Technologies

Adhesion between different materials



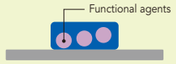
Adhesion (for a wide variety of materials) that firmly bonds different materials together

Dot coating



Small amounts of resin evenly placed to achieve high adhesive strength

Functional dots



Dot coating of functional agents that enables the sophisticated incorporation of functions

Proprietary adhesive technologies + Collaboration across glass fiber and chemicals businesses = Market development and technology refinement to enter the fields of lifestyle product materials and industrial materials

Insulation Materials Business

Vision

Utilize external resources as well as those of the Nittobo Group to open up new markets

Issues and Initiatives

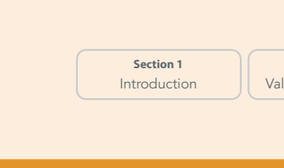
1. Take advantage of growing demand for energy-saving solutions to strengthen sales of high-end products
2. Stable supply derived from improved productivity
3. Expand recycled glass raw material resources in response to decline in number of glass bottles recovered

The Insulation Materials Business Division is responsible for the entire process from research and development to the manufacture and sales of glass wool. This business is conducted by Nittobo Group company Paramount Glass Manufacturing Co., Ltd. Glass wool, in addition to its high heat insulation properties, possesses various functions such as nonflammable and sound-absorbent qualities. It is used for a wide range of purposes, such as houses, buildings, and industrial applications, as well as in vessels and vehicles. Against the backdrop of stricter energy conservation standards and social demands for carbon neutrality, there is a growing need for improved thermal insulation properties and reduced environmental impact.

Strengths	• Ability to harness proprietary technologies to develop high-performance insulation materials and help realize energy-efficient housing
Risks	• Decline in residential new-build numbers • Threat to business performance due to rising raw material, fuel, and distribution costs
Opportunities	• Increasing demand for environmentally friendly high-end products, boosted by growing needs for energy-saving solutions

Main Products

Glass wool insulation and sound-absorbing materials for housing



Glass wool is a fiber-based insulation material made mainly from recycled glass, offering high thermal insulation, sound absorption, and nonflammable properties. It is also superior in terms of cost-effectiveness and workability, and is used in a wide range of areas such as walls, ceilings, and floors in residential buildings. With the growing popularity of highly insulated homes and energy conservation, it is widely adopted in the housing industry as an insulation material that offers both environmental performance and cost performance.

Architectural and industrial glass wool insulation and sound-absorbing materials

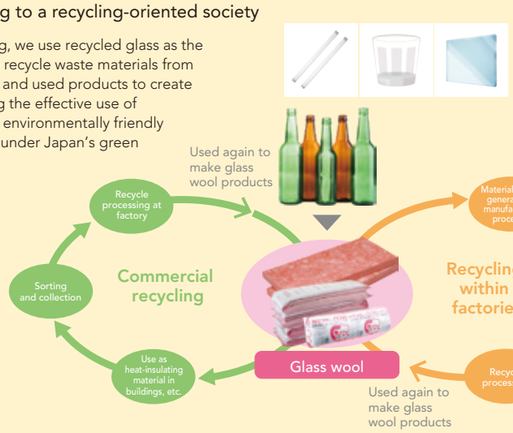


Glass wool is widely used not only in residential applications but also in buildings such as office buildings and factories, as well as in various industries. Utilizing its excellent thermal insulation, sound absorption, and nonflammable properties, it contributes to improving thermal environments and noise control. Among these applications, GC Board is an interior finishing material made by wrapping glass wool in glass cloth. It is often used in spaces that require comfort and functionality, such as machine rooms, where thermal insulation and sound absorption are essential.

Glass wool contributing to a recycling-oriented society

In glass wool manufacturing, we use recycled glass as the main raw material. We also recycle waste materials from the manufacturing process and used products to create new ones, thereby ensuring the effective use of resources. Glass wool is an environmentally friendly product that is recognized under Japan's green purchasing legislation.







Column Nittobo's Special Glass: Pioneering the Future of a Society Characterized by High-Speed, High-Capacity Communications

Nittobo's Special Glass: Pioneering the Future of a Society Characterized by High-Speed, High-Capacity Communications

As we move toward realizing a society characterized by high-speed, high-capacity communications, the standards demanded of glass fiber cloth used as an insulating material for printed circuit boards are rising. Nittobo Group is advancing research and development of high-performance Special Glass with high elasticity, low coefficient of thermal expansion, low dielectric constant and low dielectric dissipation factor, contributing to the evolution of this digital infrastructure.

Supporting Digital Society

The semiconductor market is expected to grow further with the global market predicted to expand to approximately JPY 100 trillion by 2030. With the rapid evolution and widespread adoption of generative AI technology in recent years, AI servers are also becoming more powerful and their market size is expanding.

Nittobo Group provides two main types of Special Glass through its proprietary technologies.

T-glass (high-elasticity, low thermal expansion glass) is an essential material for semiconductor package substrates supporting information and communications society, such as CPUs, GPUs, and ASICs. As semiconductor packages become increasingly sophisticated, substrates (semiconductor package substrates) are growing larger, leading to the emergence of challenges such as thermal expansion and warping. To respond to this issue, the Company's T-glass fiber cloth has been highly evaluated as an effective solution, and its adoption as a substrate has significantly increased. We will continue our research and development efforts to achieve even lower coefficient of thermal expansion and higher-speed transmission.

NE-glass was developed in the 1990s and has led the industry as the de facto standard for low-dielectric glass in recent years. In particular, it is being adopted for the motherboards of switches required for high-speed communication between chips in AI servers, and there is a sharp uptick in its demand. As the demand for higher-performance materials increases with further improvements in communication speeds, the Nittobo Group has developed NER-glass, which further upgrades the excellent low dielectric dissipation factor and low dielectric properties of NE-glass, and has begun its rollout.

	Commodity Use	Special Glass		
Properties	E-glass	T-glass	NE-glass	NER-glass
Density (g/cm³)*1	2.6	2.5	2.3	2.3
Tensile strength (GPa)*2	3.2	4.8	3.1	2.3
Tensile elasticity modulus (GPa)	75	86	64	50
Maximum elongation percentage (%)	4.8	6.1	–	-
Thermal expansion coefficient (×10⁻⁶/℃)*1	5.6	2.8	3.3	3.4
Softening point (℃)*1	844	> 1000	–	–
Dielectric constant [10 GHz]*1	6.6	5.3	4.7	4.5
Dielectric dissipation factor [1 GHz]*1	0.0060	0.0070	0.0025	0.0018
Characteristics	● Electrical insulation properties	● High tensile strength ● High tensile elasticity ● Low coefficient of thermal expansion properties	● Low dielectric constant ● Low dielectric tangent	● Low dielectric constant ● Low dielectric tangent

*1 Measured values for lumps of glass *2 Measured using glass fiber called "virgin fiber," which has less defects. Values thus obtained indicate those close to an ideal level of strength.
Note: All data listed above are examples of actually measured values and not standard ones.

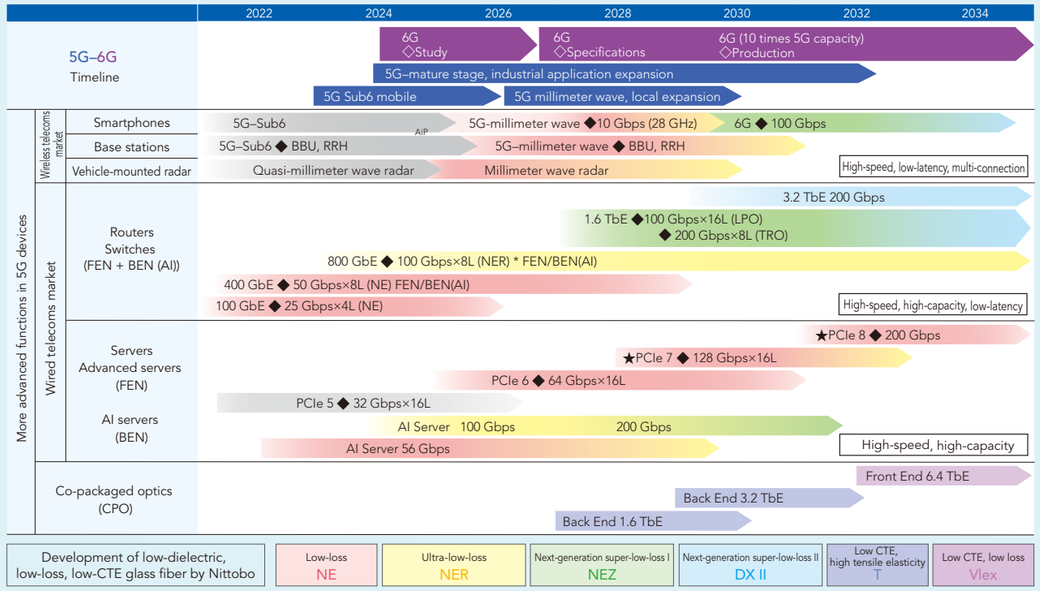
The Growing Utility of Special Glass

Special Glass is filling an expanded range of applications within the semiconductor market, particularly in the key areas of logic and memory semiconductors, where significant growth is expected. In the logic field, Special Glass is being applied in low-dielectric motherboards for AI servers and network switches, as well as in low-CTE (Coefficient of Thermal Expansion) packages for CPUs, GPUs, and ASICs. In the memory field, it is being used in low-dielectric packages for DRAMs that work alongside CPUs and GPUs, and we have also begun expanding into low-CTE packages for NAND memory.

Application		Equipment/Part	Substrate type		Required performance	Glass fiber type	
						High-end	Mid-range
Telecoms/ Infrastructure	● Mobile phone base stations ● Data center switches, routers, and servers ● AI servers	Processors Controllers	Semiconductor package substrates	CPUs/GPUs/ASICs	Low CTE	T	E
				NAND memory	Low CTE	T	E
				DDR memory	Low dielectric tangent	NE/NER	E
		Motherboards	Servers (AI, General)		Low dielectric tangent	NE/NER	E
			Switches (Core, AI)		Low dielectric tangent	NE/NER/NEZ	E
			Network cards (NIC)		Low dielectric tangent	NE/NER	E
Edge devices	● Smartphones ● Tablets ● Mobile PCs ● Edge AI devices	Processors	Semiconductor package substrates	APs/CPUs/NPUs	Low CTE	Ultra-thin T, T	Ultra-thin E
		Non-volatile memory		NAND memory	Low CTE	Ultra-thin T	Super ultra-thin E
		Volatile memory		DDR memory	Low CTE	Ultra-thin T (smartphones)	
				DDR memory	Low dielectric tangent	NE (PCs)	
		Motherboards	Motherboard substrates		Low dielectric tangent	Ultra-thin NE	Ultra-thin E
		Wireless telecoms market	RF package substrates		Low dielectric tangent	Ultra-thin NE	Ultra-thin E
	● Desktops ● Laptop PCs ● AI-PC	CPUs/Memory	Semiconductor package substrates	CPUs/GPUs/NPUs	Low CTE	T	E
		Motherboards		DDR memory	Low dielectric tangent	NE (PCs)	E
	● AR/VR ● Drones	Advanced SoCs	Semiconductor package substrates		Low CTE	T	Super ultra-thin E
Automobiles	● Electric vehicles ● ADAS	Advanced SoCs	Semiconductor package substrates		Low CTE	T	E
		Millimeter wave radar	Module boards		Low dielectric tangent	Ultra-thin NE	E

Nittobo's Technology for the Future — Responding to the 5G/6G Roadmap —

The Nittobo Group has contributed to the development of advanced information and telecoms infrastructure based on 5G and 6G in society by providing Special Glass that is used in printed wiring boards that support technological innovation in high-speed, high-capacity communications and semiconductors. We will continue our research and development activities to meet the demands of the market, leveraging our strengths as an integrated manufacturer with both spinning and weaving processes, alongside our technology for developing glass compositions with diverse properties.



Arrow color corresponds to Nittobo low-loss glass fiber materials that have been adopted or are expected to be adopted.

Source: Compiled by the Company with reference to Ministry of Internal Affairs and Communications materials

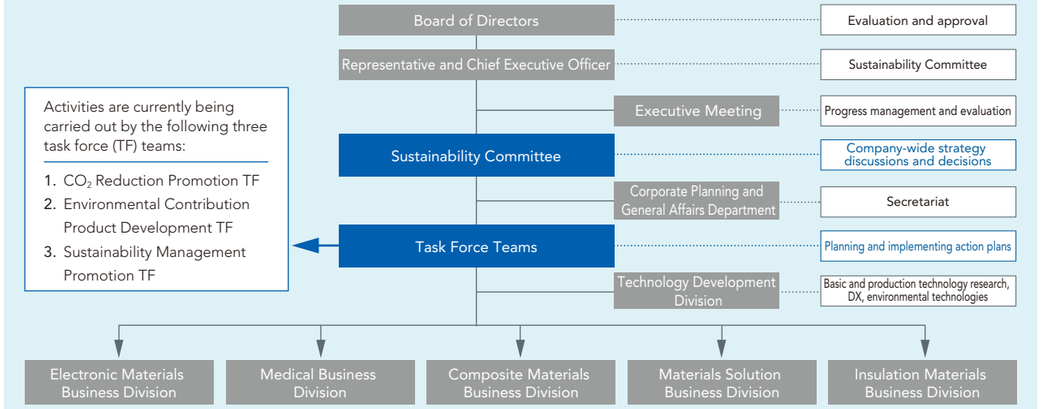
* FEN: Front-end network, BEN: Back-end network

Nittobo's Sustainability Management

The Nittobo Group is aware of its responsibilities as a corporate citizen, and addresses environmental, social, and governance (ESG) issues with the aim of "contributing to the realization of a more prosperous world," as set forth in its corporate philosophy. We contribute to society by creating value through our business activities, and we will continue to work on improvements that enhance corporate value and lead to sustainable growth.

Sustainability Management Structure

The Sustainability Committee, established in April 2021, is chaired by the Representative and Chief Executive Officer and comprises the Division General Managers of each Business Division, the Division General Manager of the Corporate Management Division, and the Division General Manager of the Technology Development Division. The Sustainability Committee meets regularly, and important matters discussed and decided are reported to the Board of Directors for discussion and resolution.



Sustainability Committee Agenda for FY2024

13th Meeting	14th Meeting	15th Meeting	16th Meeting
• Summary of FY2023 initiatives and report on planned initiatives for FY2024 • Report on CO_2 emission reduction scenarios for 2050 • Future ESG management initiatives throughout the Group	• Performance report on solar power generation system at Fukuyama Enterprise Center • Report on carbon footprint calculation • Progress report on plans to obtain ISCC PLUS certification	• Consideration of specific measures to reduce CO_2 emissions • Status of development of products using recycled glass • Progress report on the development of products that contribute to the environment	• Report on ISCC PLUS certification acquisition and operational status • Analysis of ESG evaluations and consideration of countermeasures • Report on capital investment to reduce CO_2 emissions

Discussions, Decisions and Progress from the 1st through 12th Sustainability Committee Meetings

- We revised our CO_2 emission reduction targets for FY2030, identified issues at our glass fiber production sites and are promoting activities such as considering new initiatives to reduce CO_2 emissions in the production of glass fiber, which account for the majority of our CO_2 emissions.
- We have endorsed the TCFD recommendations, and we examine, disclose and annually review the risks and opportunities posed by climate change in all of the Group's operations.
- We are preparing to calculate our carbon footprint and disclose it to the market and our customers. Internally, we are factoring in an internal carbon price (ICP) of JPY 10,000 per ton of CO_2 emitted when making capital investments and are accelerating measures to reduce CO_2 emissions.

Key Initiatives

Energy efficiency

Promotion of thorough energy-saving measures

- Improve yield and productivity, strengthen technologies, consider introduction of high-efficiency combustion equipment, etc.
- Expand measures at model factories to the entire Company as an all-employee initiative

Electrification and switching to alternative fuels

Electrification of gas furnaces and consideration of next-generation furnaces

- Implement electrification and hybridization of gas furnaces
- Conduct technical studies of new furnaces with a view to switching to alternative fuels

Introduction of renewable energy

Installation of solar power systems

- Consider and implement the introduction of solar power generation systems at factories and on idle land
- Expand use of renewable energy

Product development

Development of products that contribute to the environment

- Further strengthen and digitalize planning and development of products that contribute to the environment
- Strengthen collaboration with customers to verify product performance

Mindset reform

- Foster understanding and awareness of sustainability activities
- Increase sensitivity for identifying business opportunities

System creation

- Promote energy-saving capital investment by introducing internal carbon pricing
- Engage in active external communication to enhance corporate value

Collaboration and cooperation

- Promote collaboration with local governments and other companies

Harmonious Coexistence with Society

The Nittobo Group's corporate philosophy calls upon us to create healthy and more comfortable lifestyles. This involves developing together with all of our stakeholders and society as a whole, including shareholders, investors, local governments and regional communities, and implementing various initiatives to realize a more prosperous world.

Community Contribution

The Company serves as an official club partner of Fukushima United FC (a professional soccer club in the J3 League) with the aim of promoting the development and revitalization of Fukushima Prefecture through sports.





Support for Education

At the Fukuyama Enterprise Center and the Fukushima Enterprise Center, staff conduct "outreach classes" at local schools and also conduct factory tours for students.




A factory tour

Environmental Initiatives

The Nittobo Group has set proactively addressing environmental issues as a provision in the Nittobo Behavior Program, which in conjunction with the Nittobo Environmental Charter and the Companywide Environmental Policy, provide the framework under which the Group strives for ongoing improvement. Furthermore, in accordance with Big VISION 2030, we are working to realize a sustainable society by developing products that reduce environmental impact and contribute in the areas of the environment and energy.

Nittobo Environmental Charter

Guiding Principle

In all our corporate activities and business development efforts, we, the Nittobo Group, will contribute to handing down the irreplaceable global environment to the next generation and ensuring sustainable development by providing environmentally conscious products and services and taking the needs of the environment into consideration in our activities.

Action Guidelines

1	Environmental protection	Aim at improving the environment continuously and coexisting with the local community by striving to reduce environmental impacts in a systematic way.
2	Environmentally conscious corporate activities	Promote energy-saving, resource-saving, recycling-oriented, and environmentally conscious operations in all business domains in Japan and abroad.
3	Compliance with environmental policy and voluntary management	Conform to laws and regulations in addition to other requirements and principles as well as stipulate voluntary environmental policy and management standards and work toward achieving them.
4	Periodic audits	Conduct periodic environmental audits to verify the appropriateness of management standards and improve environmental and safety management to raise such standards.
5	Training and education	Provide all employees with thorough environmental and safety training and raise their environmental and safety awareness levels mainly by making environmental and safety policies known to them, and strive to hand down the global environment to posterity.

Established on October 1, 1998 Revised on April 1, 2023.

Companywide Environmental Policy

1	Contribution to the future of humankind and the global environment	We aim to reduce CO_2 emissions by rationalizing energy use and switching to low-carbon fuels.
2	Ensuring people's health and the safety of local communities	We manage hazardous substances appropriately and strive to reduce their discharge into the environment.
3	Toward a recycling-oriented society	We aim to achieve zero emissions through emission control and proactive adoption of recycled raw materials.
4	Compliance with laws and regulations	We comply with environment-related laws and regulations and regional agreements, and work to totally eliminate environmental risk.

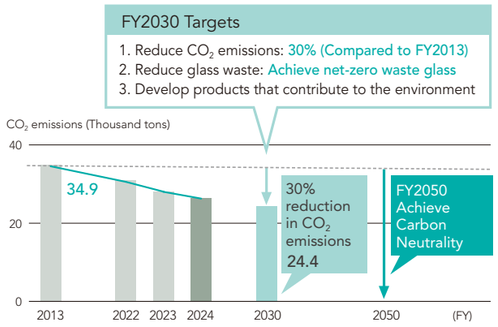
Initiatives to Reduce CO_2 Emissions

		FY2030	FY2050
1	Greenhouse Gas Emission Reduction Targets	Reduce by 30% (compared to FY2013)	Achieve net-zero
	Efficient use of energy in power supply/production facilities	Reduce energy/improve efficiency of power supply facilities	Reduce energy/improve efficiency of production facilities
	Introduce/Expand use of renewable energy	Introduce solar power generation systems Investigate introduction of other renewable energy	Introduce/Expand various types of renewable energy
	Convert energy used to melt glass/change melting method	Improve energy efficiency of melting process Investigate/Introduce CCUS	Convert to low-carbon fuels Expand use of CCUS
2	Targets for reductions in waste glass	Achieve net-zero waste glass	
3	Develop products that contribute to the environment	Promote/Expand use of renewable raw materials Reduce carbon footprint (disclose to customers)	Glass fiber using recovered glass fiber raw materials Carbon-neutral glass fiber (CN-GF)* (zero carbon footprint)

* Glass fiber that achieves net-zero CO_2 across all processes, from raw material extraction to transportation and manufacturing.

Greenhouse Gas Emission Reduction Targets

The Nittobo Group has identified "providing products and services that contribute to the environment and energy" and "adapting to climate change and a resource-recycling society" as material issues to address in the course of its business activities. We are working to reduce climate change-related environmental impact and prevent global warming by improving productivity, promoting energy-saving activities, efficiently using energy through recycling, and utilizing solar power generation. To achieve carbon neutrality by FY2050, we are promoting initiatives that use various means to advance the decarbonization of our supply chain and, in turn, broader society.



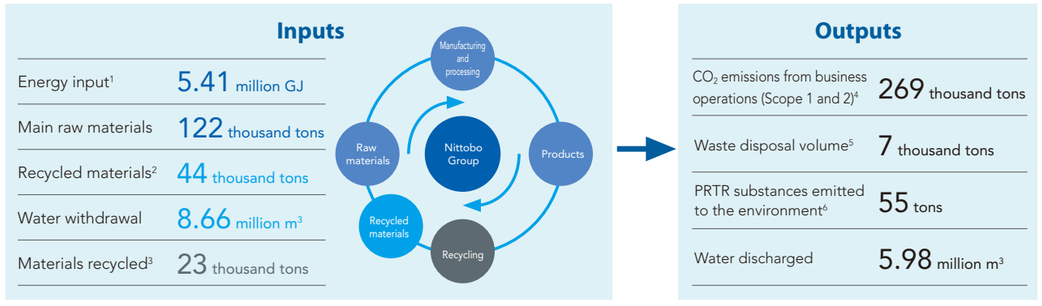
For details of our response to climate change:
https://www.nittobo.co.jp/eng/sustainability/environment/climate_change.html

Reducing the Nittobo Group's Environmental Impact

The Nittobo Group uses fuel and electricity in the manufacture of a wide range of products made from natural resources and chemical substances. As a result, the main environmental burdens include CO_2 from energy consumption, chemical materials that are emitted into the environment as unrecoverable, and waste materials. The Nittobo Group takes responsibility for conserving the environment and strives to reduce its environmental impact through the rational use of resources.

Results for FY2024

The Nittobo Group conducts an environmental conservation survey once a year among its global consolidated subsidiaries. At our glass fiber manufacturing sites, which account for the majority of the Group's CO_2 emissions, we conduct a monthly energy survey. The material balances shown in the diagram below are compiled from environmental impact data for the Nittobo Group's global consolidated companies.



Scope of Data Collection

Japan business sites: Data collected from April 2024 to March 2025
Overseas business sites: Data collected from January to December 2024

1. **Energy input:** Total energy used in the form of fuel and electricity at factories, offices and other sites.
2. **Recycled materials:** Amount of recycled resources received as raw materials by our factories from other companies.
3. **Materials recycled:** Portion of waste material generated at factories and other sites that is used effectively or recycled for reuse.
4. **CO_2 emissions from business operations:** Amount of CO_2 emitted through the consumption of energy at factories and other business sites.
5. **Waste disposal volume:** Portion of waste material generated at factories and other sites that is not used effectively and is simply incinerated, sent to landfill sites, or otherwise disposed of.
6. **PRTR substances emitted to the environment:** Amount of substances that are subject to the PRTR Act (the Act on Confirmation, Etc. of Release Amounts of Specific Chemical Substances in the Environment and Promotion of Improvements to the Management Thereof) emitted to the atmosphere or in wastewater at factories and other sites.

Setting and Operation of Internal Carbon Pricing System

In August 2022, we introduced our own internal carbon pricing (ICP) system.

We aim to accelerate measures to reduce CO_2 emissions by factoring in ICP when evaluating the benefits of capital investments.

Taking into account the cost of renewable energy currently applied by electric power companies and the Company's current CO_2 emissions, etc., we have set the ICP at JPY 10,000 per ton of CO_2 emissions. This figure will be periodically reviewed when formulating medium-term management plans.

Examples of expenses to which ICP is applied are:

- Energy-saving measures for existing equipment
- Introduction or renewal of equipment to improve energy efficiency
- Introduction of renewable energy equipment
- Conversion of equipment to use low-carbon fuels

Development of Products That Contribute to the Environment

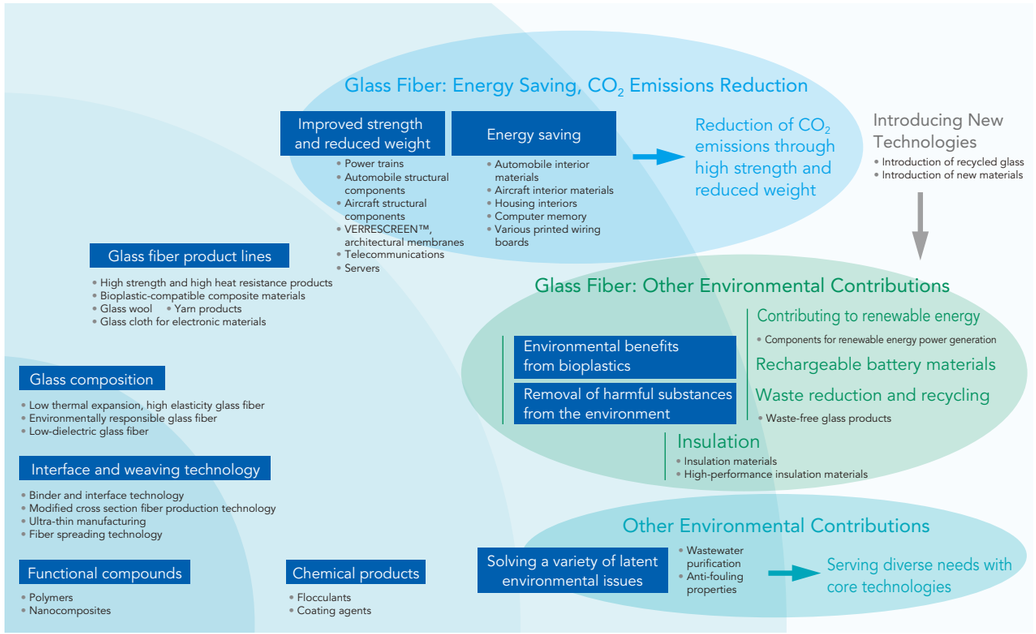
We will contribute to the reduction of environmental impact through our businesses by making full use of the Nittobo Group's technological capabilities and product knowledge.

High-Performance Insulation

Paramount Glass Manufacturing Co., Ltd., a Nittobo Group company, manufactures and sells glass wool insulation used in homes and other buildings. Insulation is a material that contributes to energy saving by preventing outdoor heat and cold from flowing into rooms and by improving the effectiveness of indoor heating and cooling.

Japan's Act on the Improvement of Energy Consumption Performance of Buildings (Building Energy Efficiency Act) was enacted in 2015 to strengthen global warming countermeasures. It has since been revised with expanded and additional provisions. As of 2022, all new residential construction must conform to the highest conventional insulation grade by 2025. Thus, there is an increasing need for insulation with improved performance, and even higher thermal insulation ratings are being established. In response, we have been developing high-performance insulation. While maintaining the inherent properties of glass wool insulation, namely fireproofing, high-durability, and low CO_2 emissions throughout the product lifecycle, we develop products with insulation performance superior to that of other materials that will meet the newly established high thermal insulation ratings. High-performance insulation can be expected to improve insulation performance by approximately 30% compared with conventional glass wool insulation.

Direction for Development of Products That Contribute to the Environment



Initiatives in Support of the TCFD Recommendations

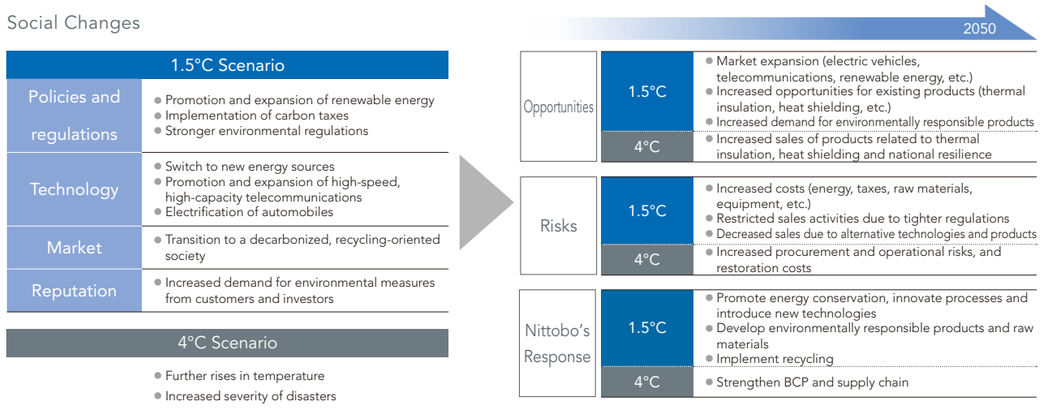
Governance

In May 2022, the Nittobo Group endorsed the recommendations of the Task Force on Climate-related Financial Disclosures (TCFD). The Sustainability Committee is responsible for identifying climate-related risks and opportunities, as well as formulating countermeasures. The activities of the Committee, including important matters which have been deliberated, are reported to the Board of Directors, where they are supervised and discussed.

Strategy (Opportunities and Risks)

With regard to the impact of all of the Group operations on climate change, we conducted an analysis looking at 2030 and 2050 with reference to scenarios drawn up by specialized organizations such as the Intergovernmental Panel on Climate Change (IPCC) and the International Energy Agency (IEA).

The risks posed by climate change can be divided into those associated with transitioning to a low-carbon society Strategy (Opportunities and Risks) (transition risks) and physical impacts (physical risks). For the two scenarios where the average global temperature rises either 1.5°C or 4°C above preindustrial levels, we identified items that are likely to have a high impact in terms of opportunities and risks. We will continue to review action plans as appropriate in response to changes in the external environment, and reflect such changes in our plans.



See below for details of scenario analysis.
https://www.nittobo.co.jp/eng/sustainability/environment/climate_change.html

Risk Management

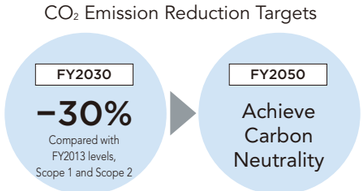
The Nittobo Group's Sustainability Committee and Risk Management Committee are responsible for the planning and progress management of CO_2 emission reduction across the Group.

In order to identify and assess the impact of climate change on the Group's operations, we identify and analyze climate change opportunities and risks, discuss them in the Sustainability Committee, and consider countermeasures.

Metrics and Targets

The Nittobo Group has identified "providing products and services that contribute to the environment and energy" and "adapting to climate change and a resource-recycling society" as material issues to address in the course of its business activities. We are working to reduce climate change-related environmental impact and prevent global warming by improving productivity, promoting energy-saving activities, efficiently using energy through recycling, and utilizing solar power generation. To achieve carbon neutrality by FY2050, we are promoting initiatives that use various means to advance the decarbonization of our supply chain and, in turn, broader society.

CO_2 Emission Reduction Targets



Fukushima Enterprise Center and Fuji Fiber Glass Obtain ISCC PLUS Certification

In 2024, the Company obtained ISCC (International Sustainability and Carbon Certification) PLUS certification, one of the international certification systems for sustainable products, at both the Fukushima Enterprise Center (Fukushima City, Fukushima) and our wholly-owned subsidiary, Fuji Fiber Glass Co., Ltd. (Moka City, Tochigi).

At the certified factories, certified products can now be handled using the mass balance method*. ISCC PLUS certification is a system that ensures raw materials derived from renewable resources, etc., are properly managed throughout the supply chain, including product manufacturing. The Company will conduct its business in accordance with the latest ISCC regulations when handling sustainable products that utilize recycled materials manufactured under the mass balance method.



Fukushima Enterprise Center


Fuji Fiber Glass

* Mass balance method:
A method in which, when products are manufactured using multiple types of raw materials (e.g., recycled or biomass-based materials together with virgin or petroleum-derived materials), a portion of the product is allocated the characteristics of the recycled or biomass-based materials in proportion to the amount of such materials input.

Nittobo Advantex Joins the "Less Micro Plastic" Project

In June 2025, Nittobo Advantex announced its participation in "Less Micro Plastic," a project operated by itochu fashion system co.,ltd. Nittobo Advantex aims to contribute to reducing marine microplastics, a leading cause of ocean pollution, by utilizing its proprietary technologies, such as interlining and functional dot coating.

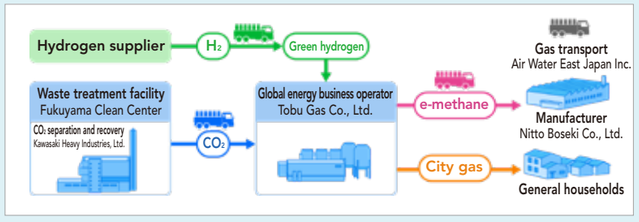
Nittobo Advantex's functional dot coating "D-ALIGN®" has the advantage of being able to be applied to materials where previously it was difficult to impart functionality. Additionally, depending on the fabric, it can prevent lint shedding during washing. Through its participation in the "Less Micro Plastic" project, it is expected to contribute to the reduction of marine microplastics.

Going forward, as part of the "D-ALIGN®" service, Nittobo Advantex will check the status of lint shedding and, upon customer request, affix the "Less Micro Plastic" logo to products meeting the certification standards. It will then process and sell these products, advancing its environmental initiative.

Koriyama City and Related Companies Launch Initiative for Exhaust Gas Reuse

In August 2025, the Company signed a collaboration agreement with Koriyama City, Fukushima Prefecture, Kawasaki Heavy Industries, Air Water East Japan, and Tobu Gas for a new initiative toward carbon neutrality. This initiative aims to establish a carbon cycle business model (see diagram) that separates and recovers CO_2 from Koriyama City's waste incineration facility exhaust gas. The CO_2 is then converted into fuel for local circulation and use. The Company will evaluate, from an industrial perspective, and utilize e-methane, which is generated by a methanation process of a portion of recovered CO_2 using green hydrogen. We envision utilizing the e-methane in the through flow boiler operated at Nittobo's Fukuyama Enterprise Center, utilizing the heat in the form of steam supplied to the factory.

We will promote this initiative in cooperation with Koriyama City and related companies in preparation for the full-scale demonstration project starting in FY2026.



Active Roles for Diverse Talent

To ensure the Nittobo Group remains the "Best Partner" for all stakeholders and continues to contribute to the realization of a more prosperous world, we believe it is important to value and harness the capabilities, thinking, and views of every employee in the Group. In line with that approach, the whole Nittobo Group strives to motivate all employees based on respect for different people, regardless of gender, age, SOGI,* nationality, disabilities, or other individual identities. Bringing about such an environment and organizational culture will create diverse value and transform our business.

Based on the Nittobo Declaration, the Nittobo Behavior Program, and the Nittobo Code of Conduct, management and the Human Resources Department work together to actively identify and resolve any issues through direct dialogue with frontline sites and employee awareness surveys. As an indicator of progress in this regard, we have retained raising the employee engagement score as one of the targets of the Medium-Term Management Plan.

* Sexual orientation and gender identity

Message from the Executive Officer Responsible for Human Resources



The future will be shaped by talent who continue to evolve.

Hiroyuki Tanida
Deputy General Manager of the Corporate Management Division

In an era marked by unexpected changes, the cycle of strategic updates is accelerating and the extent of those changes is expanding, therefore organizations must acquire skills unlike those they relied upon in the past. If the skills demanded by an organization cannot be acquired, their strategies will be nothing more than idle dreams and the Company will be unable to achieve sustainable growth. The concept of human resources has shifted to a focus on talent, and it has become a common refrain that human capital is the source of competitive advantage.

The Nittobo Group, anticipating this trend, has recognized the importance of human resources as talent and focused on individuals. While maintaining our unique culture, rooted in Tohoku, of diligence, seriousness, and a spirit of taking on challenges that has been passed down through generations, we have been developing professional talent who can fulfill their potential globally as part of making Big VISION 2030 a reality.

For such talent to play active roles, the organization must be accepting of the values and diversity of each individual. When we talk about diversity, we tend to focus on the ratio of female managers or the proportion of non-Japanese employees, but these are just one small aspect of diversity management. What we aim for is diversity in experience and skills. For this reason, we must constantly acquire new skills, update our work, and actively welcome external talent.

I am confident that these efforts will bring about organizational transformation and improved capabilities, and lead to the achievement of Big VISION 2030.

The Nittobo Group is not an organization that relies on a single superstar. Only when everyone working at Nittobo can take center stage and we unite all of our strengths can we become an organization that can compete. Everyone has their own motivation for working, but at Nittobo, our ideal is to be "an organization in which all individuals who continually enhance and transform their skills are able to shine," and this is the key to opening the door to the future.

Active Roles for Diverse Talent

Diversity Equity and Inclusion

Discussion by employees with different perspectives and personalities can lead to various ideas toward meeting diverse customer needs and providing peace of mind and confidence. Employees are the driving force that supports the Nittobo Group's sustained growth, and each member of our diverse workforce is a valuable asset. Nittobo considers it important to create an organization accepting of diversity—an organization in which all individuals can fully demonstrate their capabilities. One way of laying the foundation for such a diverse organization is to respect and promote diversity equity and inclusion.

1. Initiatives for Women's Advancement

Maintaining and promoting a workplace environment in which women play active roles is an important issue for the growth of all employees, regardless of gender. We actively promote the hiring of women, as shown by the ratio of women among new graduates we hire, which has recently reached 30%. In addition, the ratio of female managers has been steadily increasing, reaching 8.3% in FY2024. We have set a target of 10% for the future, and we are systematically training and promoting the next generation of executives.

2. Initiatives to Support Balancing Childcare and Nursing Care

It is important to create an environment where employees can balance work with childcare and nursing care responsibilities, and continue to fulfill their potential. We are implementing measures to support this balance among diverse talent.

Specifically, we have a childbirth leave system, childcare leave system, shorter working hours system, and nursing care leave systems. All eligible female employees take maternity leave and childcare leave, and many male employees take advantage of the paternity leave system when their partner gives birth. In addition, for employees working in the Tokyo metropolitan area, we offer the use of a nanny service to counter the problem of daycare waiting lists. We have also introduced a flexible working system for childcare, which can be combined with the shorter working hours system.

Most recently, we introduced a system that allows employees to take up to two weeks as paid childcare leave of absence, as well as a system that allows childcare leave to be split across two periods. These efforts resulted in 61.5% of eligible male employees taking paternity leave in FY2024. We will continue to enhance the system and further encourage male employees to take advantage of it.

3. Employment of People with Disabilities

Companies are increasingly being called upon to fulfill their responsibility to employ people with disabilities. The Company is promoting initiatives to maintain our achievement of the legally mandated employment rate for people with disabilities. As we forecast continued growth in our employee numbers in the future, we will explore various initiatives and promote activities to maintain this employment rate.

As part of our efforts to employ more people with disabilities, we strive to ensure safe working environments at our factories. We have a work support office at the Tokyo Headquarters, and we are promoting stable employment under the supervision of on-site support staff.

4. Work after the Age of 60

The decline of the working-age population is a major social issue in Japan, where the birthrate is declining and the population aging, and competition to secure talent is likely to become more intense. Against that backdrop, we will enhance programs and mechanisms that encourage employees to pursue a long career at Nittobo. We are focusing particularly on enhancing HR programs and benefits, and developing a working environment to enable employees to continue working with high levels of motivation after the standard retirement age of 60. We will also implement initiatives so that throughout their careers employees can be glad they chose Nittobo.

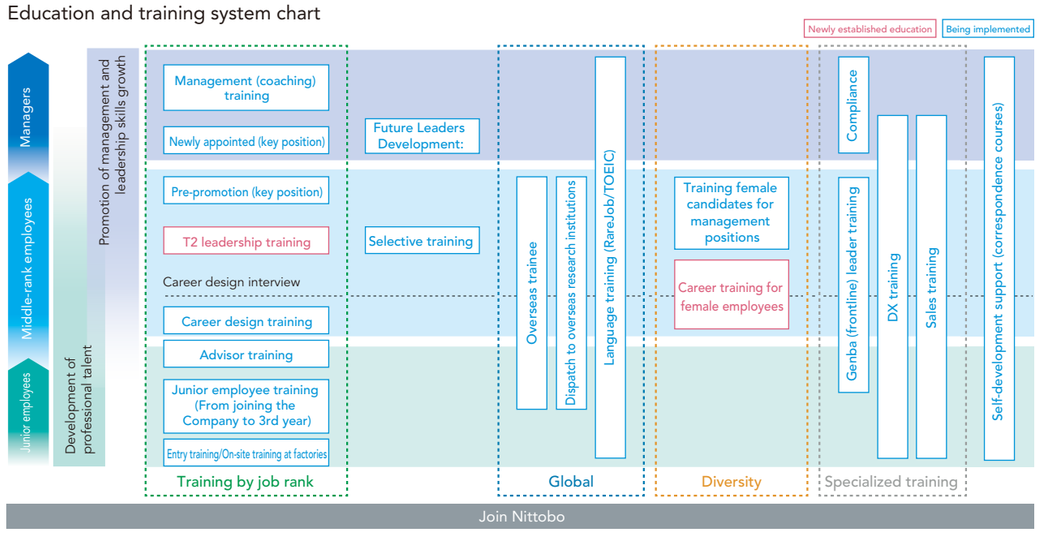
Human Resource Development

To utilize the maximum potential of the Company's diverse talent, we have been carefully implementing talent management tailored to each individual, including measures such as continuously improving both the personnel system to increase individual motivation and the training system to foster growth.

Under Big VISION 2030, we will develop next-generation leaders, global talent and professional talent. We are actively expanding management measures to promote individual motivation, self-development programs to encourage self-driven learning, and career development under the theme of fostering human resources that generate innovation, which is the foundation for business growth and driving reform. Moreover, to develop digitally capable personnel, we are conducting online digital transformation (DX) training for a broad range of participants across all Group companies, not just employees in technical roles.

We consider talent development aimed at boosting frontline capabilities to generate sustained growth to be a key measure under the Medium-Term Management Plan that began in FY2024. To that end, we are promoting the genba (frontline) leader training that we started in FY2022 and launched career training specifically for women working at manufacturing sites.

Education and training system chart



1. Development of the Next Generation of Leaders

Conventional stratified training for personnel is provided for new graduate entrants, junior employees, middle-rank employees, and newly promoted management personnel, through to division and department managers. In addition, to complement this standard training, Nittobo launched its Successor Development Programs, which are designed to match the growth stage of each employee, focusing on junior employees and middle-rank employees, in order to develop future leaders. The programs consist of intensive training for business skills and action learning based on practical issues that we need to resolve. The training in these programs promotes the development of employees' understanding of the role of senior managers and skills. In today's unprecedented management environment, developing talent who can transform business paradigms is indispensable to our business strategies. As such, Nittobo will systematically develop the leaders to drive our operations.

2. Development of Global Talent

To prepare for further globalization of the Company's business, it is essential that Nittobo fosters talent and creates a corporate culture that enables diverse individuals to fulfill their potential. We are enhancing mechanisms that support the self-directed efforts of individuals to become who they want to be, such as providing opportunities and systems for employees to engage in online English classes, and to travel overseas to study foreign languages, work at international research institutes. We are also launching an overseas trainee system that utilizes our overseas bases and will begin operation in FY2023.

Active Roles for Diverse Talent

3. Development of Professional Talent

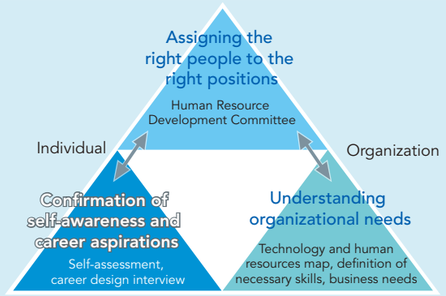
On the path to becoming a manager, Nittobo employees pursue development as specialists (professionals) in areas of their choice and work to establish their core skills. At our manufacturing sites, we are systematically promoting skills transfer through specialized education at each workplace. We have significantly expanded our subsidy system for employees seeking to obtain nationally recognized qualifications necessary for work (including a congratulatory gift of up to JPY 500,000 upon successful acquisition), and provide subsidies for distance learning, including e-learning. We foster a culture of self-driven learning and provide an environment that encourages reskilling to adapt to changing times.

4. Expansion of Management Measures to Develop Human Resources

Training courses are an important part of any human resources development policy, but it is also critical to encourage individuals to be motivated to learn, to carefully consider their career aspirations, and to implement development measures that align with their ideas and expectations. To this end, we have systematically implemented one-on-one meetings between superiors and subordinates as an important communication tool. In addition, we are implementing management measures to promote growth, such as by establishing the Human Resource Development Committee, which includes supervisors, management, and representatives from the Human Resources Department. The committee discusses future training measures and career development for employees. We have also instituted "career design training," which involves reviewing their career and thinking about their future career development, in the sixth year after joining the Company and after they have experienced a range of assignments. We have also implemented "career design interviews" to discuss with them how to achieve their goals, as well as "return-from-leave interviews" between employees returning from childcare leave and their supervisors to discuss their future careers. In addition, we are in the process of expanding our management training (coaching, etc.) for managers who will be responsible for these activities.

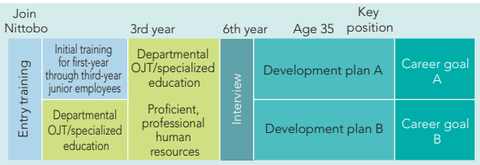
Human Resource Development Committee

The Human Resources Development Committee deliberates on how to match the needs of the organization with the unique skills and future career aspirations of each individual by utilizing self-assessments and career design interviews. The committee also considers future talent development plans and allocations.

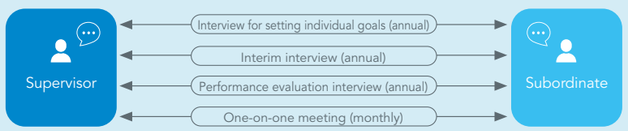


Career design interview

In their sixth year after joining the Company, when an employee has had a full range of work experience, we align understanding of their career goals. Then, they reflect on the relationship between their career goals and development plan based on the direction they intend to pursue.



Measures to Strengthen Communication




Career training for female employees

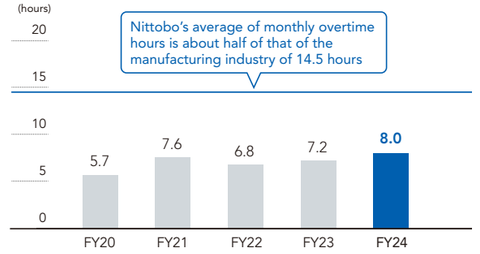

T2 leadership training

Workstyle Reform and Health Management

1. Promotion of Work-Life Balance

The average monthly overtime per employee at Nittobo in FY2024 was 8.0 hours, much lower than the manufacturing industry average. Furthermore, no employees exceeded the statutory limit on overtime hours. To provide employees with job satisfaction and enable them to demonstrate their best performance, while maintaining an appropriate work-life balance, we will continue to ensure a good working environment. Also, in Japan, which has a declining birthrate and aging population, it is important to provide a worker-friendly environment for employees who have childcare or nursing care responsibilities. We are enhancing programs that enable employees to flexibly accommodate these and other life events, such as a system of annual paid leave in hourly units and a flextime system without mandatory core hours. We aim to achieve a high level of both work-life balance and productivity improvements by reviewing rules and introducing tools.

Average Monthly Overtime Hours



2. Health Management

The Nittobo Group's corporate philosophy states that we will enhance our value to society as a corporate group by helping to create healthy and more comfortable lifestyles, and in so doing continue contributing to the realization of a more prosperous world. To continuously embody this philosophy, we believe it is essential to ensure that employees, who are the Group's driving force, are in good physical and mental health, able to work wholeheartedly and fulfill their potential.

Accordingly, we believe that maintaining and promoting each and every employee's health is a key management issue—one which we address through health management initiatives. Examples of specific initiatives include ongoing initiatives aimed at providing opportunities for physical and mental refreshment, including the introduction of a walking challenge program that uses a smartphone app, a free-of-charge use of a sports gym in cooperation with the Nittobo Health Insurance Association, and health checkups administered by post.

We have also been enhancing insurance coverage, such as by taking out comprehensive medical expense insurance to alleviate the expense burden on employees in the event of injury or sickness, and we are strengthening mental health support, such as counselling with doctors specializing in mental health.

VOICE

Comment from an Employee

I have over 10 years of experience in the research and development of in vitro diagnostic reagents and have been involved in all aspects of the R&D process since joining the Company. I find joy in bringing the Company's products to market through research and development. However, visiting hospitals where the products are actually used has made me keenly aware of the importance of maintaining a stable supply as a manufacturer.

The turning point came four years ago when I was transferred to the production department. In the Technical Department of the Production Division, our primary mission is to improve the production processes and efficiency of in vitro diagnostic reagents. Throughout my career in research and development, I have always recognized the importance of product supply as a manufacturer. I am now applying the knowledge and experience I gained in the front lines of R&D. Production sites are constantly in a race against time. Without clear answers, it is essential to quickly identify problems and generate solutions ourselves, based on communication from production sites. Recently, we have also incorporated a DX perspective and are advancing new initiatives in collaboration with the responsible departments.

In terms of work-life balance, employees can flexibly review their work duties, utilize flexible working hours, and take paid leave according to their childcare and family circumstances. Through mutual cooperation with supervisors and colleagues, we are establishing a corporate culture of mutual support.

I will continue to leverage my unique strength of having experience in both R&D and production within the Company's in vitro diagnostic reagents business. I will serve as a bridge between these two departments to ensure new products reach the market more smoothly.



Kaduho Ashizawa
NITTOBO MEDICAL CO., LTD.
General Manager of the Technical Department of the Production Division

Safety Initiatives

The Nittobo Group established Action Guidelines for safety and health in 2017, and is enhancing measures to ensure a safe and comfortable working environment that protects the physical and mental health of its employees, who are the driving force for the Group's business activities.

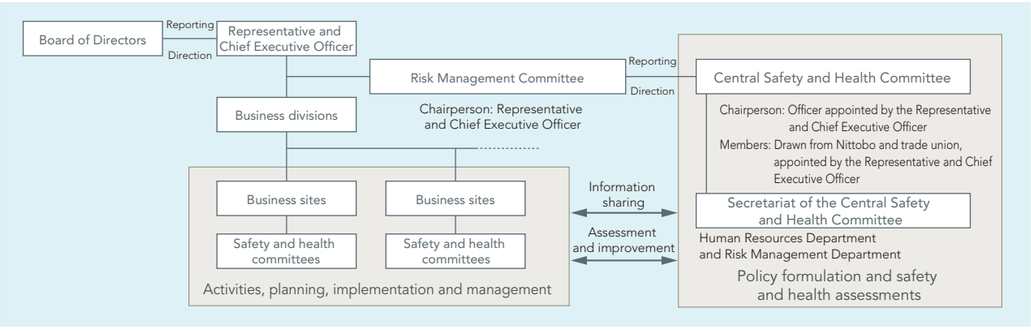
Action Guidelines

1	Establishment of a Safe Working Environment	Perform risk assessments to minimize or eliminate hazardous and harmful factors and invest the necessary management resources to maintain and improve safe and healthy working environments.
2	Promotion of Health Maintenance	Proactively assist in the maintenance and promotion of the health of employees both mentally and physically by establishing a pleasant workplace in which they can freely and willingly share and discuss their opinions, and by preventing harassment and the assignment of excessive workloads.
3	Compliance with and Management of Labor Policies	Comply with laws, regulations and other standards regarding safety and health, and set and implement internal safety policies and standards.
4	Implementation of Regular Assessments	Verify, implement and enhance safety and health management systems through regular safety and health assessments.
5	Training and Raising Awareness	Create safe and healthy working environments across the organization by thoroughly providing all employees with safety and health training and ensuring that safety and health guidelines are understood.

Established on July 1, 2017.

Management System

We established the Central Safety and Health Committee, whose members comprise representatives of Nittobo and the labor union, to discuss and formulate action policies based on the Action Guidelines, and to confirm and summarize the status of activities at business sites. The committee also submits reports to the Board of Directors and the Risk Management Committee (chaired by the Representative and Chief Executive Officer).



Major Initiatives in FY2024

Initiative	Description
Workplace Safety and Health Assessments	Business divisions and headquarters departments objectively assess safety and health activities in terms of status, behavior, and awareness. Business sites work to improve items highlighted in the assessments, which helps to standardize and enhance safety management across the Group.
Safety Improvement Month	We have designated July as Safety Improvement Month, during which we implement various initiatives throughout the Group to reaffirm the importance of occupational safety. These activities include communications regarding safety from senior management, safety awards, and safety activities in which all employees participate.
Enhanced Risk Assessments	We are enhancing our hazard prediction training and promoting efforts to raise workers' awareness of hazards and reflect this in the risk assessments. We are reviewing our work procedure manual and assessing risks at existing facilities and implementing mitigation measures starting with the highest priority facilities.
Workplace Fire Risk Assessments	We conducted fire risk assessments to identify issues at each workplace based on a check sheet developed with input from an external expert, and are systematically working to implement improvements to risk management standards at each workplace to prevent fires.
Groupwide Information Sharing	We share information including the results of safety and health assessments and fire risk assessments, safety and health best practices, and reports on disaster countermeasures in a standard Groupwide format. Doing so enables us to reflect the experience and knowledge of the Group in the activities of each business site.

Basic Stance on Human Rights

The Nittobo Group recognizes the importance of international agreements such as the Universal Declaration of Human Rights, the United Nations Global Compact and the United Nations Guiding Principles on Business and Human Rights, endorses their principles, and works to ensure respect for human rights. We recognize our responsibility to respect human rights in all our business activities.

To this end, we require officers and employees to comply with international norms, local laws and regulations regarding respect for human rights in the countries and regions where we do business. They must also refrain from any kind of harassment that violates rights, unfairly discriminates against any person in regard to gender, age, nationality, ethnicity, race, origin, religion, creed, social status, physical disability, sexual orientation or any other reason; or that gives offense to personal dignity.

In addition, we do not engage in child labor of children who are not of working age as defined by the laws and regulations of the countries and regions in which we operate, or forced labor. We provide appropriate training to all employees in order to ensure understanding of the above concepts.

Respect for Human Rights

1. Prohibition of discrimination and respect for diversity
2. Prohibition of harassment
3. Prohibition of child labor and forced labor
4. Consideration for youth, seniors, and expectant mothers, and work-life balance
5. Human resource development
6. Wages and working hours
7. Protection of the right to organize

Sustainable Procurement

To implement sustainable management, efforts on the procurement side, such as raw materials, production equipment, and energy, also play an important role. The Nittobo Group is engaged in sustainable procurement in collaboration with our suppliers. We are building robust relationships with them in order to contribute to the creation of a sustainable society through our procurement activities as well.

Implementation of a Sustainable Procurement Policy

We have formulated the Nittobo Group Sustainable Procurement Policy as a policy to share our ideas and challenges with suppliers regarding our social mission of contributing to a sustainable society, and we are conducting awareness-raising activities for our suppliers.

In addition, for suppliers who are starting new purchasing transactions with the Nittobo Group, we make sure they are familiar with the Nittobo Group Sustainable Procurement Policy and confirm that their ideas and efforts are in alignment with our thinking and initiatives. Furthermore, we also conduct surveys of our suppliers to identify circumstances that pose risks to sustainable procurement in the supply chain and take actions to resolve them. Based on survey results, we provide written feedback, including comments and improvement requests, visit our suppliers, and work to further enhance sustainable business activities. Starting this fiscal year, we are conducting a sustainable procurement survey at our overseas offices and launching initiatives similar to those in Japan.

Examples of Sustainable Procurement Initiatives

This fiscal year, we reaffirmed the significance of green procurement in order to enhance the effectiveness of the Nittobo Group Green Purchasing Guide for Office Supplies established in the previous fiscal year, designated green procurement items, and promoted the visualization of the status of implementation of green procurement.

Meanwhile, the Purchasing Department, which is responsible for managing the Nittobo Group's procurement risk, held discussions on green procurement during visits to all Group business sites and shared examples of initiatives taken at other business sites. And at the end of the fiscal year, as a conclusion to a year of green procurement activities, we conducted a survey on green procurement to identify challenges.

Green procurement of office supplies is a sustainable activity that all Nittobo Group employees can easily implement in their immediate work environment, and we believe that it will also further enhance each and every employee' awareness of their role in the Group's sustainability management.

In addition to green procurement, the Purchasing Department, through exchanges of opinions with all Group business sites on sustainable procurement risks such as conflict minerals and human rights in the supply chain, is raising awareness of supply chain management and promoting further improvement activities.

For the Nittobo Group Sustainable Procurement Policy:
https://www.nittobo.co.jp/eng/sustainability/pdf/SustainableProcurementPolicy.pdf

Directors and Executive Officers



Takeshi Miida — Director
Yasuharu Nakajima — Director
Agasa Naito — Director
Sadayoshi Fujishige — Director
Yuichi Tsuji — Director, Representative and Chairman
Hiroyuki Tada — Director, Representative and Chief Executive Officer
Takanobu Matsunaga — Director

Directors (As of June 20, 2025)

Director ●●●●●

Sadayoshi Fujishige

Career Summary

Mar. 1969 Joined Lion Fat & Oil Co., Ltd. (currently Lion Corporation)
Mar. 1996 Director, Executive General Manager of International Division of Lion Corporation
Mar. 2000 Executive Director, Executive General Manager of Home Products Sales Headquarters of Lion Corporation
Mar. 2002 Representative Director, Senior Executive Director, Responsible for Home Products Business, Executive General Manager of Home Products Sales Headquarters of Lion Corporation
Mar. 2004 Representative Director, President and CEO of Lion Corporation
Mar. 2006 Representative Director, President and CEO, Chairman of the Board of Directors, Responsible for Home Products Business of Lion Corporation
Jan. 2012 Representative Director, Chairman and CEO, Chairman of the Board of Directors of Lion Corporation
Jan. 2014 Representative Director, Chairman, Chairman of the Board of Directors of Lion Corporation
Mar. 2016 Advisor of Lion Corporation
Jun. 2020 External Director of SATO HOLDINGS CORPORATION (current) External Director of the Company (current)
Mar. 2021 Special Consultant of Lion Corporation (current)
Jun. 2021 Outside Member of the Board (Member of the Audit and Supervisory Committee) of TV Asahi Holdings Corporation (current)

Director ●●●●●

Agasa Naito

Career Summary

Oct. 2001 Registered as an attorney Joined Nagashima Ohno & Tsunematsu
May 2008 Registered as an attorney at New York State Bar Association
Jun. 2011 Joined TANABE & PARTNERS
Apr. 2013 Partner of TANABE & PARTNERS (current)
Jun. 2015 Outside Audit & Supervisory Board Member of BOOKOFF CORPORATION LIMITED
Oct. 2018 Outside Audit & Supervisory Board Member of BOOKOFF GROUP HOLDINGS LIMITED
Jun. 2020 External Director of the Company (current)
Aug. 2021 Outside Director (Audit & Supervisory Board Member) of BOOKOFF GROUP HOLDINGS LIMITED (current)
Jun. 2022 Supervisory Director of GLP J-REIT (current)
Jun. 2023 External Audit & Supervisory Member of ispace, inc. (current)

Director ●●●●●

Yasuharu Nakajima

Career Summary

Oct. 1983 Joined Tetsuzo Ota & Co. (currently Ernst & Young ShinNihon LLC)
Apr. 1987 Registered as a certified public accountant
May 1998 Partner of Showa Ota & Co. (currently Ernst & Young ShinNihon LLC)
May 2004 Senior Partner of ShinNihon & Co. (currently Ernst & Young ShinNihon LLC)
Sep. 2008 General Manager of 6th Audit Department of Ernst & Young ShinNihon LLC
Sep. 2010 Managing Director, General Manager of 2nd Audit Business Division, and General Manager of Knowledge Headquarters of Ernst & Young ShinNihon LLC
Jul. 2021 Partner of Ernst & Young ShinNihon LLC
Jun. 2022 External Director of the Company (current)
Jul. 2022 Outside Director (Audit and Supervisory Committee Member) of Business Brain Showa-Ota Inc. (current)

Director ●●●●●

Takeshi Miida

Career Summary

Apr. 1978 Joined MEIDENSHA CORPORATION
Apr. 2008 Executive Officer, General Manager in charge of the Corporate Policy Planning Group and the Corporate Policy Planning Division of MEIDENSHA CORPORATION
Apr. 2011 Managing Executive Officer, General Manager in charge of the Corporate Policy Planning Group and the Corporate Policy Planning Division of MEIDENSHA CORPORATION
Apr. 2012 Senior Managing Executive Officer and General Manager in charge of the Corporate Policy Planning Group of MEIDENSHA CORPORATION
Jun. 2012 Director, Senior Managing Executive Officer and Manager of Corporate Planning Group of MEIDENSHA CORPORATION
Apr. 2015 Representative Director and Executive Vice President of MEIDENSHA CORPORATION
Jun. 2018 Representative Director and President of MEIDENSHA CORPORATION
Jun. 2022 Representative Director & President & Executive Officer of MEIDENSHA CORPORATION
Jun. 2023 Representative Director Chairperson and Senior Officer of MEIDENSHA CORPORATION (current)
Jun. 2024 External Director of the Company (current)

Director, Representative and Chairman ●●

Yuichi Tsuji

Career Summary

Oct. 2013 Joined the Company General Manager of Accounting and Finance Department
Apr. 2014 Operating Officer and General Manager of Accounting and Finance Department
Jun. 2014 Executive Officer, General Manager of Corporate Business Planning Department and Responsible for Corporate Risk Management
Nov. 2014 Executive Officer, General Manager of Corporate Business Planning Department, Responsible for Corporate Risk Management, Human Resources Department, Information System Department, Compliance Management Department, Purchasing Department, Osaka Branch and Nagoya Branch
Jun. 2015 Director and Executive Officer (current)
Feb. 2016 Representative Executive Officer
Jun. 2016 Representative and Chief Executive Officer
Apr. 2024 Representative and Chairman (current)

Director ○●

Takanobu Matsunaga

Career Summary

Apr. 1990 Joined the Company
Apr. 2019 President and Representative Director of Paramount Glass Manufacturing Co., Ltd.
Apr. 2020 Executive Officer, Deputy General Manager of the Glass Fiber Division and President and Representative Director of Paramount Glass Manufacturing Co., Ltd.
Apr. 2021 Executive Officer, Deputy General Manager of the Glass Fiber Division, and Product Planning and Development Division, and President and Representative Director of Paramount Glass Manufacturing Co., Ltd.
Apr. 2022 Senior Executive Officer, Deputy General Manager of the Glass Fiber Division, Responsible for Sales Division, and General Manager of Product Planning and Development Division, and President and Representative Director of Paramount Glass Manufacturing Co., Ltd.
Apr. 2023 Senior Executive Officer, Vice President of the Glass Fiber Division, General Manager of Composite Material Division and the Product Planning and Development Division
Apr. 2024 Senior Executive Officer
Jun. 2024 Director (current)

Director, Representative and Chief Executive Officer ●●

Hiroyuki Tada

Career Summary

Apr. 1985 Joined the Company
Dec. 2016 General Manager of the Textile Division
Apr. 2017 Executive Officer and President of the Textile Division
Oct. 2018 Executive Officer and President of the Glass Fiber Division
Aug. 2019 Executive Officer, Responsible for Human Resources Department, Corporate Business Planning Department, General Affairs Department, Centenary Project, Corporate Communication Department, Accounting and Finance Department, Information System Department, Procurement Management Department, Osaka Branch and Nagoya Branch
Apr. 2020 Managing Executive Officer, Responsible for the Corporate Management Division and the Textile Division, General Manager of the Corporate Business Planning Division
Apr. 2021 Managing Executive Officer, Responsible for the Corporate Management Division, General Manager of the Corporate Business Planning Division, President of the Textile Division and Representative Director and President of Nittobo Advantex Co., Ltd.
Apr. 2022 Managing Executive Officer, Responsible for the Corporate Management Division, General Manager of the Corporate Business Planning Division, General Manager of Purchasing Department, President of the Textile Division and Representative Director and President of Nittobo Advantex Co., Ltd.
Apr. 2023 Representative Executive Officer and Senior Managing Executive Officer, Responsible for the Corporate Management Division and the Textile Division and General Manager of the Corporate Business Planning Division
Apr. 2024 Representative and Chief Executive Officer (current)
Jun. 2024 Director (current)

● External Director
● Directors who meet the Company's criteria for independence and for whom the Company has submitted an "Independent Directors/Auditors Notification" to the Tokyo Stock Exchange
○ Director not concurrently serving as an Executive Officer
● Member of Nomination Committee
● Member of Remuneration Committee
● Member of Audit Committee

Executive Officers (As of July 1, 2025)

Representative and Chairman	Yuichi Tsuji*	
Representative and Chief Executive Officer	Hiroyuki Tada*	CEO
Managing Executive Officers	Hisanobu Hayashi	Division General Manager of the Electronic Materials Business Division
	Yasushi Okahisa	Division General Manager of the Corporate Management Division; Responsible for the Secretary Office, the Corporate Planning and General Affairs Department, and the Purchasing Department
Senior executive officer	Akimasa Kajita	Deputy Division General Manager of the Electronic Materials Business Division; Head of the Yarn Unit; Chairman and Representative Director of Nitto Glass Fiber Manufacturing Co., Ltd.
	Hiroki Kajikawa	Deputy Division General Manager of the Corporate Management Division; Responsible for the Corporate Communication Department, the Accounting and Finance Department, and the Information Systems Department; General Manager of the Corporate Communication Department
Executive officer	Katsuya Hatanaka	Deputy Division General Manager of the Medical Business Division; Chairman CEO of Nittobo America Inc.
	Masaki Ito	Division General Manager of the Composite Materials Business Division; General Manager of the Administration and Planning Department of the Composite Materials Business Division
	Koichi Nakamura	Division General Manager of the Technology Development Division
	Masao Fukushima	Division General Manager of the Medical Business Division; President and Representative Director of NITTOBO MEDICAL CO., LTD.
	Hiroyuki Tanida	Deputy Division General Manager of the Corporate Management Division; Responsible for the Human Resources Department, the Risk Management Department, the Legal Department, the Fukushima Enterprise Center, the Fukuyama Enterprise Center, the Osaka Branch, and the Nagoya Branch; General Manager of the Risk Management Department
	Shotaro Yasuda	Division General Manager of the Materials Solution Business Division; Head of the Industrial Fabric Unit
	Takehiro Mikami	Deputy Division General Manager of the Electronic Materials Business Division; Head of the PWB Materials Business Unit; General Manager of the PWB Materials Marketing Department; Chairman of Nittobo Taiwan Co., Ltd.

* Concurrently serving as Director

Message from External Directors



Sadayoshi Fujishige External Director

Evolution of the Management Structure through the Business Division System

In 2024, the Company launched a five-business division structure (six business segments, including other businesses), and each division now integrates sales, production, and product development and production technology development directly linked to its business.

This structure allows the Company to respond quickly to market changes and customer requests. In addition, profit responsibilities have become clearer for each business division, and the speed at which issues are recognized and promoted within each business division has improved. Significantly delegating authority to the heads of business units in business operations and allowing them to experience the results of their own decision-making has led to the development of top executives ready to lead the next generation.

As a result, operating profit reached a record high in FY2024. This was made possible by the efforts of each business division to achieve its respective business responsibilities under the strong leadership of top management. I would like to emphasize that this is also the result of our decisive and continuous capital investment over the past few years, particularly in the Special Glass and medical fields.

We are continuing to make decisive capital investments with our sights set on the future growth in demand. As an external director, I have occasionally advised the Company on its medium-term management plan, focusing on its transformation into a market-sensitive company. From my perspective, the Company seems to be strongly technology-oriented.

To counterbalance that tendency, I have consistently emphasized the importance of having a keen sensitivity to market fluctuations and valuing deep dialogue with customers, expressing my inclination towards becoming a market-sensitive company at every opportunity.

The transition to a business division structure where sales, production, and research are managed as an integrated entity has made it possible for us to engage in ongoing, in-depth dialogues with our customers and further enhance our ability to quickly identify business opportunities. This has led to a transformation into a more market-sensitive company.

As an external director, I would like to engage in ongoing dialogue with the management team and on the front lines (sales, production, development, staff, etc.) to support the Company's further growth as a market-sensitive company, in addition to its strength in unique technological development capabilities.

Going forward, I believe that developing new businesses that will become our third pillar, following the Special Glass-Related Business and Medical Business, will be an important management challenge.

As the Chairperson of the Nomination Committee

Regarding the composition of the Board of Directors, efforts are being made to ensure diversity in areas such as management, legal affairs, finance, and human resources, including by appointing more female directors. The Board also places importance on having at least half of the directors be external directors for governance purposes, with a view to promoting more active discussions from diverse perspectives.

To develop next-generation management human resources, it is important to select leaders for management with diverse experience in business operations; outstanding results and accomplishments; acute conceptual, decision-making, and execution skills; and, most importantly, people skills.

We must not lose sight of the need to improve our human resources and remuneration plan in order to appoint top leadership talent from both inside and outside the Company, and to enhance our succession planning.



Agasa Naito External Director

Thoughts on Promoting Diversity and Inclusion

At Nittobo, promoting diversity and inclusion (D&I) is positioned as one of the most important management challenges for achieving sustainable growth and enhancing corporate value.

Our overall goal is to make Nittobo's Big VISION 2030 a reality and become a corporate group that continues to create Global No. 1 Niche Businesses that contribute to the Environment & Energy, Digital Society, and Health, Security & Safety, which are essential parts of building a sustainable society. To achieve this, we must foster an environment where employees with diverse values and personalities can freely exchange opinions and generate new innovations. To flexibly and accurately respond to the rapidly changing external environment and global market needs, an organization where human resources with diverse backgrounds can demonstrate their strengths is vital. Amid the inevitable decline in the working-age population due to an aging society with a declining birthrate, we need to create a workplace environment where diverse talent can fulfill their potential in order to secure and retain talented human resources over the long term.

Nittobo is working to further promote D&I as one of the pillars of its human resources strategy in its Medium-Term Management Plan. Among these pillars, promoting initiatives for women's advancement is a key theme that we are focusing on, and I personally have a strong interest in it as well. However, to truly achieve D&I, diversity must be addressed in all aspects, not just gender. This includes attributes such as nationality and age, as well as providing diverse ways of working and career paths tailored to different stages of life. To achieve this, the management team needs to continuously communicate the significance of D&I and embed and disseminate it as part of our corporate culture, rather than merely designing formal systems and setting numerical targets. We will continue to promote effective initiatives aimed at creating an organization where diverse human resources can fulfill their potential in a variety of ways.

Evaluation and Challenges of Nittobo's Governance System

Under the structure of a company with a nomination committee, etc., Nittobo has adopted a governance system based on the so-called "monitoring model." The Board of Directors is comprised of seven directors, with four independent external directors with expertise in different fields such as management, finance, accounting, and legal affairs constituting a majority of the Board. Independent external directors also comprise a majority of the Nomination, Audit, and Remuneration Committees. This system ensures a supervisory function from an independent and objective standpoint with regard to business execution. We have also established institutional and environmental systems that support appropriate risk-taking, thereby enhancing medium- to long-term corporate value and achieving sustainable growth.

Additionally, efforts are being made on the operational side to enhance the effectiveness of the system, such as providing and sharing information through briefings before Board of Directors meetings. During my attendance at Board of Directors meetings and various committees, I have been able to observe how each member utilizes their own knowledge and experience, and lively and meaningful discussions take place.

The effectiveness of the Board of Directors is evaluated annually through a questionnaire survey that is conducted and analyzed by external lawyers, and it has generally received high ratings. I feel that issues pointed out in the evaluation results are being addressed as quickly and accurately as possible.

In terms of future challenges, important themes include developing systems for nurturing the next generation of management human resources and further diversifying core human resources in fields such as international perspectives and DX. I will continue to engage in discussions and deliberations with a view to further strengthening governance.

Message from External Directors



Yasuharu Nakajima External Director

As the Chairperson of the Audit Committee

In the corporate society to which we belong, the corporate view that "the purpose of a company is to maximize profits and stock prices" is often asserted. This is absolutely true, and it is also our mission at Nittobo.

At the same time, I have the impression that Nittobo has many employees with high loyalty to the Company. Recently, some Group companies have shown remarkable improvements in engagement.

When employee engagement is high, employees feel more loyal to the Company, their level of contribution to the Company increases, and sustainable growth and medium- to long-term corporate value can be expected. Nittobo must utilize these irreplaceable human assets as an engine for growth.

However, there is a significant pitfall here that is easy to fall into with just one wrong step. Japanese-style corporate scandals often stem from a combination of a company-first mindset, profit-maximization behavior, and a tendency for blindly pursuing business without regard for the company's social responsibilities and purpose. It goes without saying that scandals greatly damage corporate value. From my experience as a certified public accountant, I know that there are many cases in Japanese corporate society where loyalty, which should be a virtue, ends up backfiring. Avoiding such situations at all costs is the primary mission of the Audit Committee.

Although companies aim to maximize profits, they must recognize that this objective is changing. New images of what a corporation should be are emerging, such as those that emphasize ESG (environment, society, and corporate governance), and there is a growing demand for the disclosure of non-financial information.

Here, I will quote from the famous book "Practical Study" by Kazuo Inamori, which is also my spiritual guide.

"Fundamentally, a capitalist society is not a society where anything goes in pursuit of profit. It is a society built on the premise that all participants must uphold social justice, and it is only with strict morality can the system function properly." "All companies need to have high ethical standards and practice fair and transparent management." In short, morality is a prerequisite.

I would like everyone working in the corporate world to always keep the following three points in mind as fundamental premises necessary for living in this world. (1) Ethics prevail over profit, (2) Never cheat, (3) False reporting is never acceptable.

These are obvious points, but they are easy to forget. Premise (3) is a part of premise (2), but the principles of proper disclosure and reporting in the financial market in which a corporation is situated carry a very heavy responsibility, so I have deliberately listed them separately.

In order to link the high level of loyalty among Nittobo employees to corporate growth, it is essential for the management to align employee motivation and values with the Company's goals, encourage them to work enthusiastically, and promote increased creativity and productivity. This is also the responsibility of the Audit Committee. I will strive daily to provide every possible support with the aim that the Company can contribute to society indefinitely, to promote management based on strong bonds (engagement) between employees and Nittobo and their ethical standards. I will always keep in mind the importance of making ethical judgments to ensure that corporate common sense does not become societal nonsense.



Takeshi Miida External Director

Changes after Completing the First Year of the Medium-Term Management Plan

Under our current Medium-Term Management Plan, the next four years are positioned as crucial in determining whether we can achieve our long-term vision of "Establishing a unique position throughout our product development and deepening our relationships with global clients." To achieve this, we must further strengthen our product development capabilities and ensure that we realize returns on our current investments.

This is my first year as an external director, but I have two impressions of Nittobo based on candid observations from an outside perspective. First, as expected, due to its long history, there are product businesses focusing on resolving past issues, although this is taking time. However, I was pleasantly surprised to see that challenging initiatives in new domains such as the highly competitive medical field have been producing results beyond expectations.

In the first half of this year, responding to the rapid changes in the electronic materials market was a major theme, but in the second half, themes such as M&A with a medium- to long-term view also became agenda items. I feel that I was able to gain a good understanding of the intentions of the Company's executives this year.

While there is always a need for portfolio transformation, I believe the most important aspect is developing our human resources.

In that sense, dialogue with the Company's executives is very important and communication with the subsidiary members through on-site inspections is an effective tool.

The business environment in which Nittobo operates is highly dynamic, so how we balance our brand power, price competitiveness, and product development capabilities will greatly influence our future. At Nittobo, the allocation of resources such as investment and human resources, as well as the timing and speed of investments that accurately capture market changes, are extremely important. The Board of Directors has established a foundation for communication that enables external directors to engage in in-depth discussions with the Company's executives on these matters.

As the Chairperson of the Remuneration Committee

The key points in designing a remuneration plan are "setting targets in line with management strategies such as long-term vision and Medium-Term Management Plan" and "appropriateness of remuneration and rationality of incentives." As a committee, we must fulfill our responsibility for transparency and accountability regarding remuneration. The degree of achievement of the Medium-Term Management Plan is an important indicator in the remuneration plan, but how to evaluate activities toward the long-term vision remains an issue. Although product development is important in the long term, its success also depends on market trends. Therefore, we need to refine our remuneration plan based on evaluations from various perspectives, such as risk tolerance and how to encourage forward-looking investments.

Ensuring transparency in the management team remuneration decision-making process is vital. This will allow us to fulfill our accountability to stakeholders, clarify the reasons for setting management team remuneration, and explain these reasons to shareholders as necessary. Through these activities, the Remuneration Committee aims to fulfill its role in strengthening the corporate governance structure and enhancing corporate transparency and accountability.

Corporate Governance

The Nittobo Group constantly updates its corporate governance system with the aim of creating a fair and highly transparent management organization in order to conduct business activities with emphasis on trust from our shareholders, investors and other stakeholders.

Timeline of Corporate Governance Enhancements

(Year)	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Corporate governance system	Company with a board of directors			Company with a nomination committee, etc. (from June 2014)											
Committees	—			Nomination Committee (Chair: External director)											
				Remuneration Committee (Chair: External director)											
				Audit Committee (Chair: Internal director)	Audit Committee (Chair: External director)										
Directors	7	6		8				7	6	7					
Of which, external directors	1	1	2	3	4 (at least half)			4 (at least half)	3 (at least half)	4 (at least half)					
Of which, women										1					
Operation of Board of Directors					Briefings before Board of Directors meetings										
							Self-evaluation of the effectiveness of the Board of Directors								
Revisions to systems, regulations							Formulation of Executive Meeting regulations								
											Introduction of performance-linked stock remuneration plan				

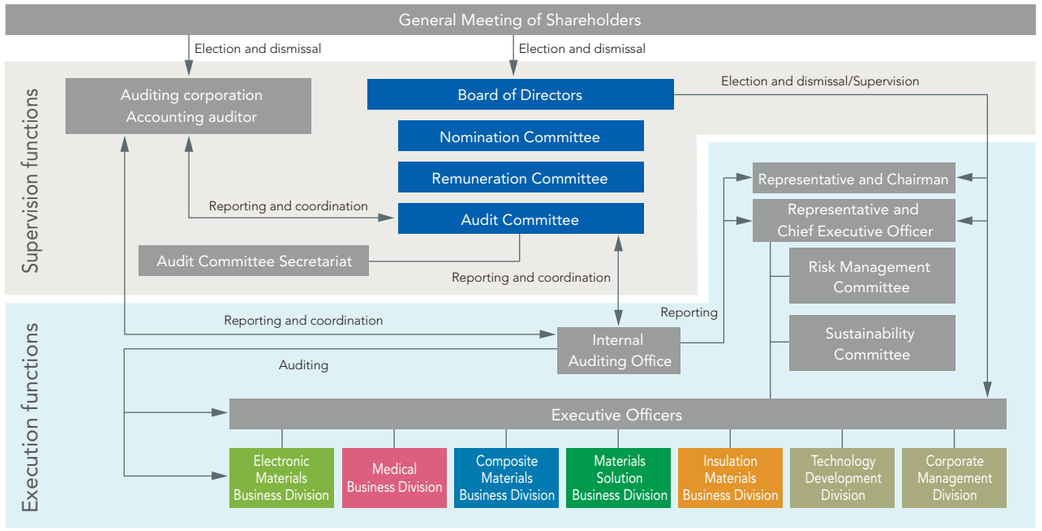
Overview of Corporate Governance System and Reasons for Employing the System

The Company introduced an executive officer system in June 2003 and other systemic measures to maximize the effectiveness of consolidated management by invigorating the Board of Directors and speeding up decision making. Since June 2008, management has been conducted with an emphasis on the clarification of the functions and roles of management and business execution.

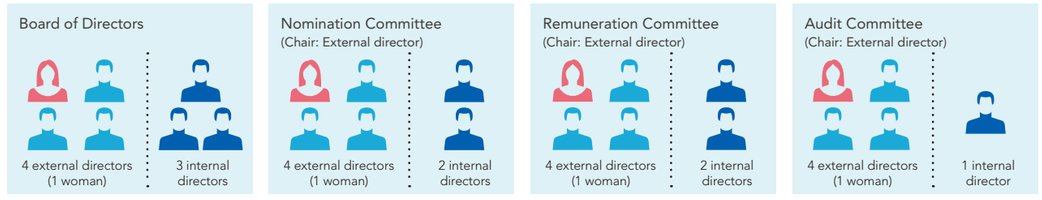
In addition, with the approval of the Ordinary General Meeting of Shareholders on June 26, 2014, Nittobo adopted the structure of a company with a nomination committee, etc. By doing so, the Company further clarified the separation of supervision from execution and aims to strengthen supervisory functions, ensure highly transparent management, execute business swiftly, and increase the flexibility of management. Furthermore, the establishment of a system that enables it to more precisely meet the expectations of stakeholders, such as customers, shareholders, suppliers, and employees, will enable the Company to further enhance its corporate value.

Corporate Governance System

Diagram of Corporate Governance System



Composition of Board of Directors and Three Committees



Board of Directors

Regarding the Board of Directors, the Company's basic policy is that it shall comprise internal directors, who are appointed from within the Company and have capabilities and knowledge of each business and management in general, and at least the same number of external directors, who are able to express opinions that can enhance governance from the perspectives of diverse stakeholders and society. This balance enables appropriate supervision of the Company's business execution by the Board of Directors as a whole. Furthermore, the Board of Directors shall have no more than 12 members, as specified in its Articles of Incorporation. At the time of publication of this report, the Board of Directors comprises seven members, of whom four are external directors.

The Board of Directors is responsible for supervision of business execution—it selects the directors that make up the Nomination, Remuneration and Audit Committees; elects and dismisses executive officers and delegates business to them; approves matters related to operations that have an impact on basic management policy such as medium-term management plans and annual budgets; and approves matters that could have a significant impact on Group management such as investments exceeding a certain amount. The term of office of directors is one year, and subject to approval at the Ordinary General Meeting of Shareholders held annually. The Company's executive officers are determined by the Board of Directors. People suitable for the execution of the Company's business are elected from the perspective of enhancement of business execution and corporate value based on the principle of assigning the right people to the right positions.

Corporate Governance

Nomination, Remuneration and Audit Committees

Under the structure of a company with a nomination committee, etc., the Company has established a Nomination Committee, a Remuneration Committee, and an Audit Committee. The roles of each committee and an overview of committee members as of the date of the publication of this report are as follows.

Nomination Committee

The Nomination Committee determines the content of proposals (regarding the election and dismissal of directors) to be submitted to the General Meeting of Shareholders. The committee comprises six directors (four of whom are external directors) and is chaired by External Director Sadayoshi Fujishige. Meetings of the Nomination Committee are held as needed. The Nomination Committee selects the Company's director candidates who are believed to be the most suitable based on factors such as their character and insight.

Remuneration Committee

The Remuneration Committee determines the content of individual remuneration, etc., of directors and executive officers. It comprises six directors (four of whom are external directors) and is chaired by External Director Takeshi Miida. Meetings of the Remuneration Committee are held as needed.

Audit Committee

The Audit Committee's responsibilities include auditing the state of execution of duties of directors and executive officers, and the preparation of audit reports. Working with the accounting auditors and the Audit Office, audits are conducted based on audit plans approved by the Audit Committee. The committee comprises five directors (four of whom are external directors) and is chaired by External Director Yasuharu Nakajima. The Audit Committee Secretariat has been established to assist with the duties of the Audit Committee. In principle, the Audit Committee holds meetings at least once a month.

Execution of Business

As of the date of the publication of this report, the Company has 13 executive officers, of whom one has been selected as representative and chairman and one as representative and chief executive officer.

An Executive Meeting has been established as an organ to deliberate matters related to the execution of business delegated by the Board of Directors, and it endeavors to ensure efficient business execution by holding meetings twice a month in principle.

Self-Evaluation of the Effectiveness of the Board of Directors

In order to verify whether the functions expected of the Board of Directors are being adequately fulfilled and facilitate the improvement of those functions, the Company performs yearly evaluations of the Board of Directors.

The Company performs those evaluations using a questionnaire format, and uses external attorneys from the standpoint of enhancing the independence and objectivity of evaluations in the supervision of the content of evaluations and the analysis and assessment of the results thereof.

Based on those results, the Board of Directors analyzes and assesses the composition, operation, supervision, support system, dialogue with shareholders and other aspects of the Board of Directors.

Process for Self-Evaluation of Effectiveness

Survey content overseen by outside attorneys
Survey outcomes analyzed and evaluated by outside attorneys
→ Analysis and evaluation by the Board of Directors based on evaluation results from outside attorneys
→ Discussion based on evaluation results to identify issues
→ Initiatives to address issues

Based on the self-evaluation results for FY2023, the Company performed the following initiatives in FY2024.

In the survey summarizing FY2024, the average evaluation score for most items was four or more out of five, indicating that the effectiveness of the Board of Directors was highly evaluated by both internal and external directors of the Company. However, opinions in the following vein were issued concerning topics to be addressed in the future. There is a need to continue furthering discussions on areas such as ensuring diversity in promoting core human resources, etc., and on executive succession planning.

The Company will continue to make improvements to ensure the Board of Directors performs supervisory functions even more effectively.

Initiatives in FY2024

- The Company received reports from executive officers on the progress and challenges of the medium-term management plan, and conducted multifaceted discussions and deliberations, including responses to changes in market and competitive environments.
- The Company received reports on the status of initiatives relating to sustainability activities and information security measures, and conducted discussions with the aim of further promoting those initiatives.
- Opportunities for communication between external directors and executive officers were enhanced, allowing for shared understanding of key issues and deeper discussions on management strategies.

Director Skill Matrix and Meeting Attendance (FY2024)

The knowledge, experience, and abilities of directors that are considered important to the Company from the perspective of corporate management are defined as business management, global business, sales, marketing, technology, research and development, finance, accounting, legal affairs, risk management, human resources, labor relations, and human resources development. We ensure that the Board of Directors is composed of people with appropriate knowledge and abundant experience in each field.

(Note 1) The table below does not represent all knowledge, experience, etc. possessed by each person.
(Note 2) 1. The attendance records of Hiroyuki Tada and Takeshi Miida at meetings of the Board of Directors and each committee cover the period from their appointments on June 20, 2024, through March 31, 2025.
2. The attendance records of Takanobu Matsunaga at meetings of the Board of Directors and the Audit Committee cover the period from his appointment on June 20, 2024, through March 31, 2025.

Name		Attendance/Number of meetings				Business Management	Global Business	Sales/Marketing	Technology/R&D	Finance/Accounting	Legal/Risk Management	Human Resources/ Labor Relations/ Human Resource Development
		Board of Directors Meetings	Nomination Committee Meetings	Remuneration Committee Meetings	Audit Committee Meetings							
Yuichi Tsuji		13/13	5/5	3/3		●	●			●	●	
Hiroyuki Tada		10/10	4/4	2/2		●	●	●			●	
Takanobu Matsunaga		10/10			10/10	●		●	●			
Sadayoshi Fujishige	External Director	13/13	5/5	3/3	13/13	●	●	●				
Agasa Naito	External Director	13/13	5/5	3/3	13/13		●				●	
Yasuharu Nakajima	External Director	13/13	5/5	3/3	12/13					●	●	
Takeshi Miida	External Director	10/10	4/4	2/2	10/10	●	●	●			●	●

Corporate Governance

Officer Remuneration

Director and Executive Officer Remuneration

Matters pertaining to the policy for determining the amount of director and executive officer remuneration, etc., and the method of calculation thereof

In accordance with the provisions of the Companies Act relating to a company with a nomination committee, etc. and with reference to objective data, etc. from external consultants, the Company's policy for determining the individual remuneration of directors and executive officers is decided by the Remuneration Committee, of which external directors comprise the majority and which is chaired by an external director.

Furthermore, the policy is designed to strongly incentivize directors and executive officers to achieve performance targets while ensuring transparency and rationality, to foster awareness of management from a medium- to long-term perspective, and to align interests with those of stakeholders including shareholders and investors. The performance indicators selected for calculating remuneration also reflect this approach.

Remuneration Weighting by Position

Position	Composition of officer remuneration (assuming performance-linked remuneration at 100%)				
	Basic remuneration	Performance-linked remuneration			
		Variable remuneration			Stock remuneration
		Company performance	Individual performance		
			Division performance	Individual evaluation	
Representative and Chairman, Representative and Chief Executive Officer, and other representative executive officers	60.0%	25.0%	—	—	15.0%
Senior managing executive officers and managing executive officers	60.0%	17.5%	5.0%	2.5%	15.0%
Senior and other executive officers	65.0%	12.5%	5.0%	7.5%	10.0%
Directors not concurrently serving as executive officers (excluding external directors)	71.0%	14.5%	14.5%		—
External directors	100%	—	—	—	—

(i) Overview of the overall policy for individual remuneration, etc. for Directors and Executive Officers

The Company's director and executive officer remuneration (excluding external directors) consists of basic remuneration (fixed remuneration) determined by position and variable remuneration (performance-linked remuneration) that varies within a range of 0% to 150% according to the level of achievement of Company performance targets for the previous fiscal year and individual performance targets based on the operating profit of the division(s) under the officer's charge and individual evaluation. In addition, executive officers receive stock remuneration (performance-linked remuneration), the level of which is linked to achievement of the performance targets of the Company's medium-term management plan.

In view of their role in supervising management from an independent and objective perspective, external directors do not receive variable remuneration or stock remuneration.

(ii) Policy related to content of basic remuneration

The amount of basic remuneration for each officer is determined by resolution of the Remuneration Committee based on a table that defines the fixed amount for each position.

(iii) Policy related to content of performance indicators for performance-linked remuneration

Variable remuneration

Variable remuneration (performance-linked remuneration) is calculated based on the achievement rate for each fiscal year. When officers resign prior to the end of the fiscal year, they are paid an amount prorated according to their tenure (excluding directors).

Performance Indicators and Percentage Reflection in Performance-linked Remuneration

Company performance	Operating profit (JPY 10 billion baseline)	Operating profit (change year on year)	ROE (change year on year)	
	40.0%	30.0%	30.0%	
Individual performance	Operating profit of division(s) under charge (budget variance rate)		Individual evaluation	
	Senior managing executive officers and managing executive officers	67.0%	Senior managing executive officers and managing executive officers	33.0%
	Senior and other executive officers	40.0%	Senior and other executive officers	60.0%

Reason for Selecting Performance Indicators Related to Variable Remuneration

● **Company performance**

The selected indicators are operating profit as a measure of earning power and enhanced growth potential, and ROE as a measure of improvement in capital efficiency.

● **Individual performance**

The selected indicators are the operating profit of the division(s) under the officer's charge and the individual evaluation rating, which helps to clarify the responsibility for performance of each officer (excluding external directors). These indicators also provide a basis for evaluation based on perspectives and factors other than company performance. Ratings for individual evaluations are based on the outcomes of activities and their contribution to business performance.

Calculation Rules for Variable Remuneration

Company performance component = Basic remuneration for the preceding fiscal year × Company performance coefficient for each position × Level of achievement of Company performance for the preceding fiscal year

Individual performance component = Basic remuneration for the preceding fiscal year × Individual performance coefficient for each position × Level of achievement of individual performance for the preceding fiscal year and the result of the individual evaluation

Variable remuneration is determined by combining the Company performance and individual performance components calculated as shown above.

Note that targets and actual values for Company performance in fiscal 2024 used in the calculation of performance-linked remuneration in fiscal 2023 were as follows.

Operating profit (amount) Target: JPY 10,000 million; Actual: JPY 8,387 million
Operating profit (year on year) Target: JPY 4,880 million; Actual: JPY 8,387 million
ROE (year on year) Target: 4.0% (after adjustment); Actual: 6.1% (after adjustment)
Note: Quantitative and qualitative individual performance targets were set, with achievement rates ranging between 101% to 150%.

Stock remuneration

(1) Eligibility for the performance-linked stock remuneration plan (hereinafter "the Plan")

Persons eligible for the Plan are the executive officers (including those concurrently serving as directors).

(2) Structure of the Plan and granting of points

(i) Structure of the Plan

Remuneration under the Plan will consist of the following:

● **Fixed component**

The fixed component is granted according to the position held during the tenure.

● **Performance-linked component**

The performance-linked component is granted according to the level of achievement of the performance targets of the medium-term management plan.

The level of achievement of the performance targets is evaluated each fiscal year and reflected in the remuneration for the tenure.
In the current medium-term management plan, the Company has set the amount of consolidated operating profit in each fiscal year set at the time of plan formulation as a performance indicator. Operating profit was chosen because it is a representative measure of medium- to long-term improvement in business performance. The Company's operating profit target for FY2024 was JPY 12.5 billion, and actual performance was JPY 16.4 billion. The target for FY2025 is JPY 16.9 billion.

(ii) Point calculation method

- Points corresponding to the fixed component and the performance-linked component are calculated and granted for the tenure of executive officers.
- Points granted for the respective tenure will accumulate until retirement, and the number of accumulated points will be used as "1 point = 1 share" to calculate the number of the Company's shares to be granted.

(Point formula)

Base points corresponding to the position during the tenure for which points are granted (Note 1) × (1 + Performance-linked coefficient (Note 2))

(Note 1) Base points

Position	Points
Representative and Chairman	2,360
Representative and Chief Executive Officer	2,360
Senior managing executive officer	1,219
Managing executive officer	903
Senior executive officer	452
Executive officer	409

(Note 2) Performance-linked coefficient

(The indicator for calculating the performance-linked coefficient shall be operating profit)
Performance-linked coefficient = Actual amount for the relevant period ÷ Target amount for the relevant period as specified in the medium-term management plan (rounded down to the second decimal place). The upper limit of the coefficient is 1.5, and if it is less than 0.5, it is set to 0.

(iii) Calculation method in case of position change during the tenure

If there is a change in position during the tenure, points will be prorated based on the number of months, according to the respective position.

(iv) Calculation method in case of resignation of an officer during the tenure

If an officer resigns during the tenure, points will be calculated based on his/her term of office.

(Formula)

Points calculated according to (ii) above × Tenure as an officer ÷ 12

(v) Point granting date

The points for the tenure are granted on the date of the first meeting of the Remuneration Committee after the conclusion of the Ordinary General Meeting of Shareholders for the fiscal year under review.

(3) Number of shares and amount of cash to be granted or paid

(i) In the case of the resignation of an officer due to reasons other than personal reasons

· Shares

The number of shares is calculated as "1 point = 1 share" using the following formula.

(Formula)

Number of shares = Number of points accumulated until the vesting date (including the points granted on the vesting date. Hereinafter referred to as "Number of Points Held") × 70% (any fraction less than one unit of shares shall be rounded down)

· Cash

The cash amount is calculated using the following formula.

(Formula)

Cash amount = (Number of Points Held – Number of shares calculated according to the formula for "Shares" above) × Market value of the shares as of the vesting date

(ii) In the case of the resignation of an officer due to personal reasons

Shares are granted based on the Number of Points Held, where "1 point = 1 share."

(iii) In the case of the death of an officer for whom remuneration is to be paid

In the case of the death of an officer for whom remuneration is to be paid, the right to receive a monetary payment as a survivor's benefit will be acquired by the surviving family members of such officer if they meet the requirements set forth in the Regulations of Directors' Stock Benefits established by the Remuneration Committee.
The survivor benefit amount shall be the monetary amount calculated using the following formula.

(Formula)

Survivor benefit amount = Number of Points Held by a deceased officer for whom remuneration is to be paid × Market value of the shares as of the date of death(Note)

(Note) The market price of the shares used in the plan shall be the closing price of the shares on the principal stock exchange on which the shares are listed on the date when the market price of the shares needs to be calculated. If the closing price is not published on such date, the market price of the shares shall be calculated retroactively to the most recent date for which a closing price is available.

(4) Important note

The maximum number of shares corresponding to the points awarded for each position as stipulated in Article 34, paragraph 1, item (iii)-(a)1 of the Corporation Tax Act is: 5,900 points for the Representative and Chairman and Representative and Chief Executive Officer, 3,047 points for senior managing executive officers, 2,257 points for managing executive officers, 1,130 points for senior executive officers and 1,022 points for executive officers.

Reason for Appointment of External Directors

Name	Reason for nomination as candidate for external director and overview of expected role
Sadayoshi Fujishige	Mr. Sadayoshi Fujishige has a wealth of experience and a high level of insight as a top manager of a listed company. Based on this, he actively makes statements at meetings of the Board of Directors, the Nomination Committee, the Remuneration Committee, and the Audit Committee. In this way, he plays a role expected as an external director, including providing accurate advice to management, contributing to decision making by the Board of Directors, and supervising business execution.
Agasa Naito	Ms. Agasa Naito has a wealth of experience and a high level of insight as an expert in law and as an outside officer of another company. Based on this, she actively makes statements at meetings of the Board of Directors, the Nomination Committee, the Remuneration Committee, and the Audit Committee. In this way, she plays a role expected as an external director, including providing accurate advice to management, contributing to decision making by the Board of Directors, and supervising business execution.
Yasuharu Nakajima	Mr. Yasuharu Nakajima has a wealth of experience and a high level of insight as an expert in accounting and as an outside officer of another company. Based on this, he actively makes statements at meetings of the Board of Directors, the Nomination Committee, the Remuneration Committee, and the Audit Committee. In this way, he plays a role expected as an external director, including providing accurate advice to management, contributing to decision making by the Board of Directors, and supervising business execution.
Takeshi Miida	Mr. Takeshi Miida has a wealth of experience and a high level of insight as a top manager of a listed company. Based on this, he actively makes statements at meetings of the Board of Directors, the Nomination Committee, the Remuneration Committee, and the Audit Committee. In this way, he plays a role expected as an external director, including providing accurate advice to management, contributing to decision making by the Board of Directors, and supervising business execution.

Cross-Shareholdings

Policy Regarding Cross-Shareholdings

The Company holds shares of listed companies under its policy on cross-shareholding in cases where it deems such shareholding to be beneficial for building favorable trading relationships with important trading partners of the Nittobo Group in fields including sales, fuel and raw material procurement, and finance, smoothly advancing business activities of the Nittobo Group, and maintaining and enhancing corporate value in the Nittobo Group. However, from the perspective of maintaining and improving the Group's corporate value, the Company will sell shares that are deemed to be of little holding significance, while paying attention to the effect on the market, and so forth.

Method for Assessing Rationale for Holding Shares

In holding shares in listed companies, the Company reviews the holding policy for listed shares individually at a meeting of the Board of Directors on a periodic basis, comprehensively considering qualitative factors such as sales and procurement, the existence of any technological cooperation or joint capital investment, whether there are any joint ventures under way, financing, the maintenance and establishment of good business relationships, and a quantitative evaluation that compares the total return on investment, calculated by considering dividend yield and business profit, against the cost of capital, as well as the comprehensively considered policy on cross-shareholdings.

Exercise of Voting Rights Associated with Cross-Shareholdings

The Company will appropriately exercises its voting rights associated with cross-shareholdings after consideration of whether such issuing company has established an effective governance structure and is making appropriate decisions to improve its corporate value over the medium to long term, as well as whether such shareholdings will contribute to enhancing the Group's corporate value.

Changes in Cross-Shareholdings

			FY2020	FY2021	FY2022	FY2023	FY2024
Sold during the period	Unlisted stocks	Number of stocks	1	1	—	—	3
		Amount (JPY million)	24	0	—	—	26
	Listed stocks	Number of stocks	4	—	1	—	6
		Amount (JPY million)	8,665	—	2,516	—	1,957
	Total	Number of stocks	5	1	1	—	9
		Amount (JPY million)	8,689	0	2,516	—	1,983
Capitalization at end of period (including unlisted stocks)		Number of stocks	49	47	47	47	43
		Amount (JPY million)	16,625	15,243	13,643	22,595	20,549

Action to implement management that is conscious of cost of capital and stock price

The Group will steadily work toward achieving the Medium-Term Management Plan and it will continue to undertake initiatives to achieve sustainable growth and improvement of corporate value over the medium to long term by improving ROE and maintaining an appropriate PER level. In the Medium-Term Management Plan, Company-wide quantitative targets call for ROE of 8% or more (a level that consistently exceeds the cost of capital) and ROIC at a level that exceeds WACC. The ROE for FY2024 was 10.4%, compared to the quantitative target of 8% set in the Medium-Term Management Plan. We will continue to strive for management with a keen awareness of cost of capital and stock price.

Dialogue with Stakeholders

The Corporate Communication Officer is responsible for general dialogue with shareholders, and the Corporate Communication Department is the internal department responsible for providing assistance thereto. The Company endeavors to create opportunities for dialogue with shareholders that contribute to the sustained growth and enhancement of medium- to long-term corporate value of the Company centered on executive officers. Furthermore, departments involved in IR activities centered on the Corporate Communication Department work to coordinate among departments on a daily basis.

We will continue to endeavor to disclose information in a timely and appropriate manner to enable the Nittobo Group's customers, shareholders and investors to accurately recognize and judge the actual state of the Nittobo Group. In order to achieve this, we comply with laws, regulations and securities exchange rules on disclosure, and are engaged in the creation and operation of an appropriate disclosure system.

❶ We publish items specified for disclosure in Japanese and foreign laws, regulations and securities exchange rules in business reports, annual securities reports and communications to shareholders, and announce them in securities exchange communication systems and press releases.
❷ In principle, we publish information being disclosed on the Group's website, and also endeavor to provide fairer and broader disclosure.
❸ We conduct briefings for analysts and institutional investors promptly after the announcement of quarterly financial results.
❹ We publish an integrated report that incorporates financial and non-financial information, including CSR, in order to provide a better understanding of the Group's medium- to long-term value creation framework.

Opinions and concerns, etc., about the Company that are learned through dialogue with shareholders and investors are collated by the Corporate Communication Department, reported to the Corporate Communication Officer, the Executive Meeting, and the Board of Directors on a quarterly basis, and appropriately provided as feedback to management executives. Adequate care is given regarding insider information in individual inquiries and dialogue with the Nittobo Group, and explanations are limited to information that has already been published and facts that are well known.

Compliance and Risk Management

The Nittobo Group places utmost priority on compliance with domestic and foreign laws, regulations and social norms, as well as on appropriate management and prevention of risks in areas related to its business.

Compliance

The Nittobo Group is aware that it is a public instrument that operates for the benefit of society. It naturally complies with laws and regulations as well as with social norms, and strives to remain a company that is trusted by society. This is the starting point for the Company's existence, a position that must be unconditionally observed so long as this Company is a member of society.

All officers and employees understand the necessity of practicing compliance, and the Company will always conduct fair and just business activities with a spirit of complying with the law and a firm sense of ethics.

The Nittobo Behavior Program and the Nittobo Code of Conduct

The Nittobo Behavior Program and the Nittobo Code of Conduct have been enacted as standards for all officers and employees of the Nittobo Group, to ensure that the Nittobo Group will always be a company that is trusted by society.

To ensure that these standards are firmly established as the basis for the Company's daily business activities, and that they are imbued in the core being of all officers and employees, all officers including senior management will strive to lead by example as they continuously and proactively promote awareness through education and training programs.

Internal Whistleblowing Contact

The Nittobo Group has established a whistleblowing program, the Corporate Ethics Helpline, to prevent and detect violations of laws and regulations at an early stage and to take appropriate corrective actions.

In cases where a violation of laws, regulations or internal rules, improper conduct, or other such problem occurs at work but resolution at the business site is unlikely or an employee is unable to consult with workplace colleagues or superiors, any Nittobo Group employee can make reports to and consult with the helpline with peace of mind.

The reporting and consultation office is available not only as an in-company hotline but also externally through an outside law firm, which enables anonymous reporting and consultation. The privacy and confidentiality of whistleblowers is protected, and measures are taken to prevent prejudicial treatment as a result of making a report. The reporting and consultation office also ensures independence from the executive side by handling reports and consultations that may involve directors and officers under the Audit Committee's supervision. The Company is committed to creating a system that whistleblowers can use with greater peace of mind by, for example, responding to reports and consultations in good faith, confirming the facts and taking corrective measures, and reporting the results of the investigation to the whistleblower.

For details about compliance initiatives:
https://www.nittobo.co.jp/eng/sustainability/governance/compliance.html

Risk Management

The Nittobo Group endeavors to appropriately manage the risks to which its businesses are exposed in order to improve its corporate value and fulfill its social responsibilities. We have established a system to identify, evaluate and take countermeasures against risks, which have become more diverse and complex as globalization has progressed, and are working to prevent them from materializing and to reduce losses in the event of a crisis.

Risk Management System

The Nittobo Group's Risk Management Committee is chaired by the Representative and Chief Executive Officer. It draws on the basic risk management policy and risk management system stipulated in the Group's Rules for Risk Management to appropriately manage the risk exposure of the Group's businesses and prevent problems from materializing. Furthermore, the Company has also strengthened its legal risk management system by establishing a system that enables it to receive various kinds of guidance as the occasion demands, such as opinions and advice from corporate attorneys and specialist lawyers regarding corporate management and daily operations.

Information Management

The Nittobo Group discloses information on its business activities in a timely and appropriate manner to enhance corporate transparency. In addition, we have established rules for the management of trade secrets, document management, personal information handling and information system management, and manage such information after specifying management methods and managers to ensure we appropriately manage the information in our possession. We also conduct regular self-inspections, monitoring and workplace training.

Information Security

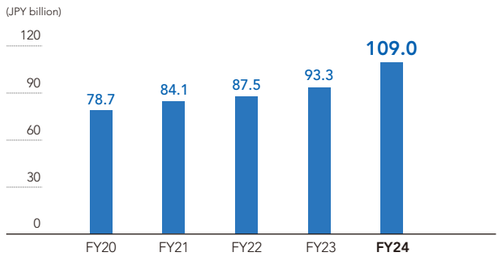
With the recent acceleration of information networking, the risk of confidential information leaks and cyberattacks is increasing. For this reason, the Nittobo Group, led by the head of IT, is strengthening preventive measures against these threats and measures to be taken in the event of an incident. By implementing measures from various perspectives, we aim to build a robust system and ensure and further enhance information security.

For an overview of information management education:
https://www.nittobo.co.jp/eng/sustainability/governance/risk_management.html

Financial and Non-Financial Highlights

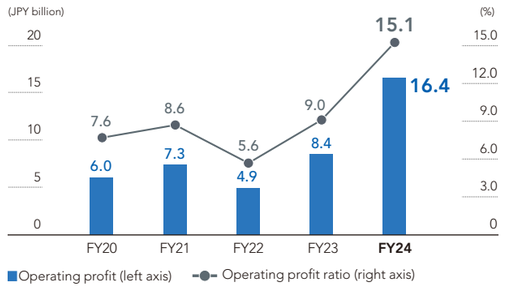
Financial Value

Net sales



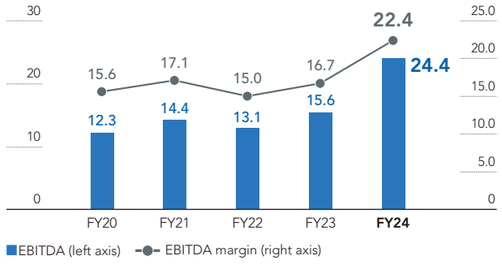
Net sales increased due to strong sales of Special Glass (a high-value-added product) in the Electronic Materials Business and the weak yen.

Operating Profit / Operating Profit Ratio



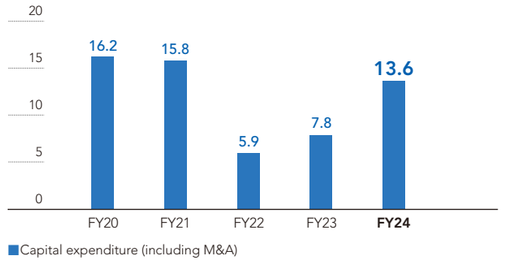
In the Electronic Materials Business, although there was an increase in foundation reinforcement expenses and depreciation, operating profit increased due to strong sales of Special Glass.

EBITDA / EBITDA Margin



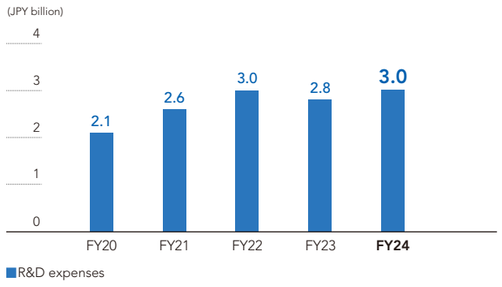
In the Electronic Materials Business, sales increased due to strong sales of Special Glass (a high-value-added product) and EBITDA margin increased along with an increase in EBITDA.

Capital expenditure (including M&A)



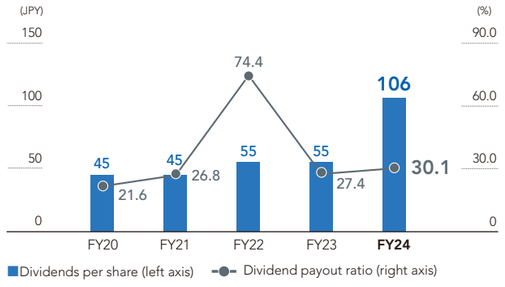
Nittobo made capital investments in growth areas, including for increasing production capacity, mainly for Special Glass yarn in the Electronic Materials Business.

R&D expenses



We enhanced the development of high-performance glass fiber for electronic materials, for which demand is expected to grow further, and new products.

Dividends / Dividend Payout Ratio



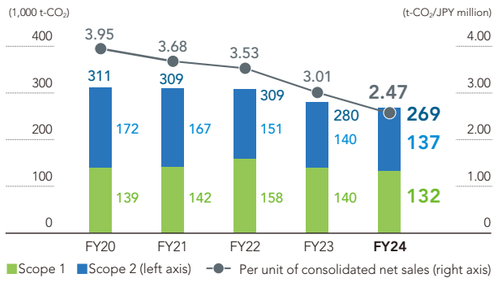
Our basic policy is to pay the annual dividend of not less than JPY 55.00 per share, targeting a dividend payout ratio of 30% of steady earnings. An interim dividend of JPY 27.50 and the year-end dividend was JPY 78.50 distributed in FY2024, for a total of JPY 106.00.

Financial and Non-Financial Highlights

Non-Financial Value

CO_2 Emissions from Business Operations

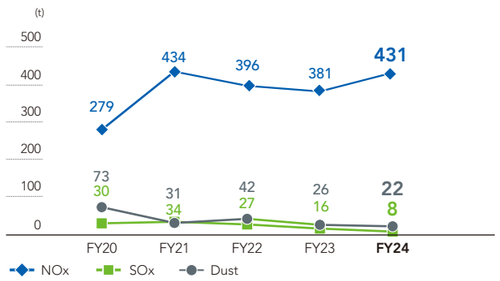
CO_2 emissions | 22.9% reduction in total amount (vs FY2013)*



* FY2013 CO_2 emissions: 349,000 t-CO_2; per unit of consolidated net sales: 4.10 t-CO_2/JPY million

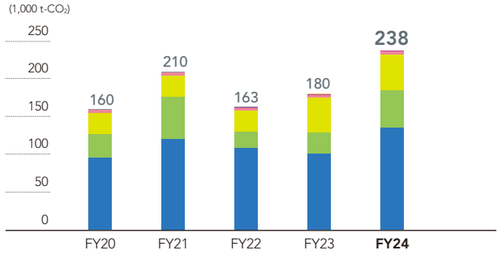
Air Pollutant Emissions

Air pollutant emissions | 53.4% reduction in total amount (vs FY2013)*



* FY2013 NOx emissions: 852 tons; SOx emissions: 43 tons; dust emissions: 95 tons

Greenhouse Gas Emissions in the Value Chain (Scope 3*)



* Scope 3: Indirect emissions outside of Scope 1 and Scope 2 (emissions of other companies related to business operations)

FY2024 Scope 3 Category Details (t-CO_2)

1: Purchased goods and services.....135,397
2: Capital goods.....50,113
3: Fuel- and energy-related activities not included in Scope 1 or Scope 2.....47,171
4: Upstream transportation and distribution.....3,864
5: Waste generated in operations.....683
6: Business travel.....358
7: Employee commuting.....311

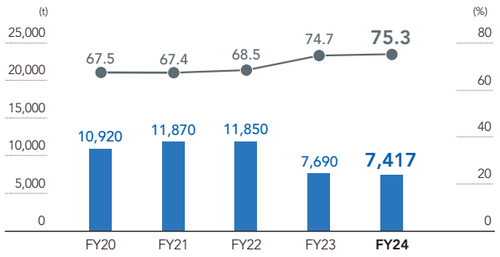
Scope 3 Reduction Initiatives

- Improve transportation efficiency in cooperation with suppliers and transporters
- Reduce CO_2 emissions through glass waste recycling

For details of third-party assurance:
https://www.nittobo.co.jp/eng/sustainability/pdf/IndependentAssuranceStatement.pdf

Waste Disposal Volume / Recycling Rate

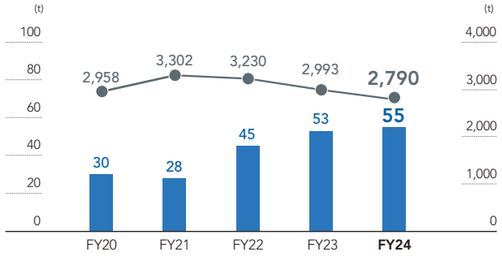
Recycling rate* | 75.3% (FY2024)



Recycling rate = Materials recycled ÷ (Waste disposal volume + Materials recycled)

PRTR Substances Emitted and Handled

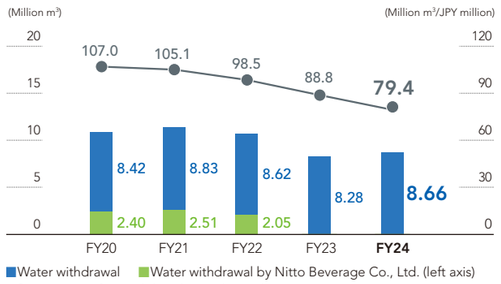
PRTR substances emitted | 57.0% reduction (vs FY2013)*



* PRTR substances emitted in FY2013: 128 t, PRTR substances handled: 2,881 t

Water Withdrawal

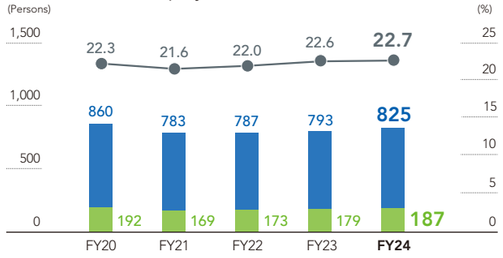
Water withdrawal | 23.4% decrease per unit of sales (vs FY2013)*



* FY2013 water withdrawal: 8.83 million m³; water withdrawal by Nitto Beverage Co., Ltd.: 2.27 million m³; per unit of consolidated net sales: 103.7 m³/JPY million

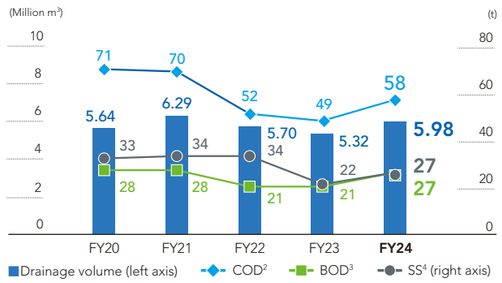
Note: Nitto Beverage was removed from the scope of consolidation in FY2022 and is therefore not included in the water withdrawal results for FY2023 or FY2024.

Total Employees (Non-Consolidated) / Ratio of Female Employees



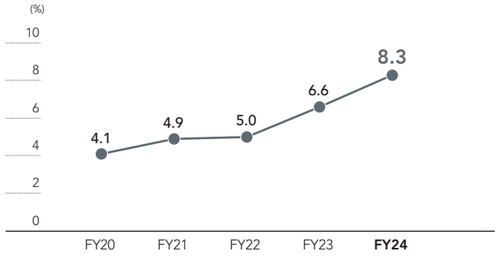
Water Drainage Volume / Water Pollutants

Water pollutants | 43.7% reduction in total amount (vs FY2013)[1]



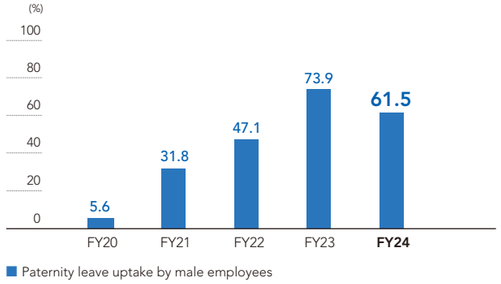
1. FY2013 COD: 116 t, BOD: 35 t, SS: 48 t, Drainage volume: 5.31 million m³
2. COD (chemical oxygen demand): Amount of oxygen required to chemically oxidize organic matter in water with an oxidant
3. BOD (biochemical oxygen demand): Amount of oxygen required for organisms to decompose organic matter in water
4. SS (suspended solids): Amount of insoluble substances in water

Ratio of Female Employees in Management Positions (Non-Consolidated)



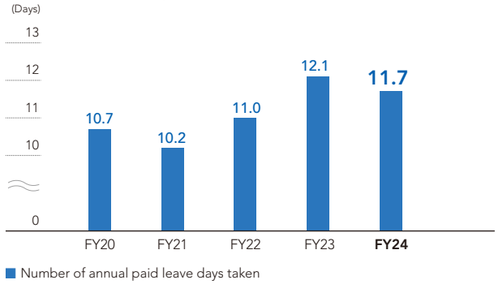
For details of third-party assurance:
https://www.nittobo.co.jp/eng/sustainability/pdf/IndependentAssuranceStatement.pdf

Paternity Leave Uptake by Male Employees



For details of third-party assurance:
https://www.nittobo.co.jp/eng/sustainability/pdf/IndependentAssuranceStatement.pdf

Number of Annual Paid Leave Days Taken (Non-Consolidated)



11-Year Financial Data

(JPY million)

	FY2014	FY2015	FY2016	FY2017	FY2018	FY2019	FY2020	FY2021	FY2022	FY2023	**FY2024**
Operating results											
Net sales	90,223	86,199	83,324	84,526	82,292	85,722	78,727	84,051	87,529	93,253	**109,035**
Operating profit	8,885	10,893	11,148	10,837	8,198	8,160	5,964	7,268	4,880	8,387	**16,445**
Ordinary profit	8,658	10,974	11,396	11,071	8,934	8,202	6,274	8,065	6,067	9,752	**17,568**
Profit attributable to owners of parent	4,588	5,598	7,479	10,253	7,984	5,771	8,100	6,519	2,772	7,296	**12,837**
EBITDA[1]	13,578	15,497	15,278	15,001	12,279	13,548	12,296	14,351	13,088	15,553	**24,386**
Depreciation	4,692	4,603	4,130	4,164	4,080	5,388	6,332	7,082	8,208	7,166	**7,941**
Capital expenditure	4,814	4,422	2,043	7,024	11,949	16,717	14,709	15,820	5,919	7,762	**13,617**
R&D expenses	1,290	1,255	1,429	1,590	1,477	1,698	2,097	2,577	2,950	2,825	**2,983**
Cash flows											
Net cash provided by (used in) operating activities	14,745	16,677	11,034	7,791	5,317	10,614	7,815	6,965	7,643	5,057	**19,121**
Net cash provided by (used in) investing activities	(5,318)	(12,294)	(2,271)	2,524	(11,389)	(16,917)	(1,867)	(17,305)	1,979	(7,896)	**(11,418)**
Net cash provided by (used in) financing activities	(4,388)	(2,276)	(8,826)	(9,467)	3,948	12,628	1,862	(1,530)	(7,249)	4,301	**(3,277)**
Free cash flows	9,426	4,382	8,762	10,316	(6,071)	(6,302)	5,948	(10,339)	9,622	(2,838)	**7,702**
Financial position (At end of fiscal year)											
Total assets[2]	145,956	142,719	140,064	147,160	151,000	172,824	184,652	187,289	185,585	212,112	**223,105**
Net assets	73,228	75,455	78,281	87,548	90,740	98,704	104,389	110,638	108,948	123,707	**135,829**
Interest-bearing debt	34,016	32,673	27,358	20,225	25,589	40,315	44,162	46,211	46,061	53,139	**52,749**
Stock information[3]											
Number of shares issued (Thousand shares)	247,677	247,677	199,677	39,935	39,935	39,935	39,935	39,935	37,723	37,723	**37,723**
Earnings per share (EPS) (JPY)[4]	23.03	28.1	37.71	263.97	205.76	148.73	208.77	168.04	73.94	200.39	**352.61**
Book-value per share (BPS) (JPY)[4]	363.32	374.50	396.00	2,232.43	2,312.06	2,413.82	2,555.06	2,727.58	2,850.72	3,246.26	**3,563.47**
Dividends per share (JPY)[5]	5.00	6.00	7.00	26.00	40.00	45.00	45.00	45.00	55.00	55.00	**106.00**
Dividend payout ratio (%)[6]	21.7	21.4	18.6	15.2	19.4	30.3	21.6	26.8	74.4	27.4	**30.1**
Management indicators											
Operating profit ratio (%)	9.8	12.6	13.4	12.8	10.0	9.5	7.6	8.6	5.6	9.0	**15.1**
EBITDA margin (%)	15.0	18.0	18.3	17.7	14.9	15.8	15.6	17.1	15.0	16.7	**22.4**
Return on equity (ROE) (%)[7]	6.8	7.6	9.8	12.5	9.1	6.3	8.4	6.4	2.6	6.6	**10.4**
Equity ratio (%)[8]	49.6	52.3	55.3	58.9	59.4	54.2	53.7	56.5	55.9	55.7	**58.1**
Debt-to-equity ratio (Times)[9]	0.47	0.44	0.35	0.23	0.29	0.43	0.45	0.44	0.44	0.45	**0.41**

1. EBITDA = Operating profit + Depreciation
2. Partial Amendments to Accounting Standard for Tax Effect Accounting (ASBJ Guidance No. 28, February 16, 2018) and other accounting standards have been applied from the beginning of fiscal 2018, and the figures for fiscal years 2014 through 2017 are shown after retroactive application of such accounting standards.
3. On October 1, 2017, a reverse stock split was implemented at a ratio of 1 share for every 5 shares of common stock.
4. Earnings per share is calculated based on the average number of issued shares during the period excluding treasury stock, and book value per share is calculated based on the number of shares issued at the end of the period excluding treasury stock.
5. The fiscal 2017 dividend of JPY 26 per share is the sum of the interim dividend of JPY 3.5 before the reverse stock split and the year-end dividend of JPY 22.5 after the reverse stock split. The dividend per share for fiscal 2017, converted on a post-reverse stock split basis, would be JPY 40.
6. The consolidated dividend payout ratio is calculated based on the average number of issued shares outstanding during the period excluding treasury stock.
7. Return on equity (ROE) = Profit attributable to owners of the parent ÷ (Net assets [average of beginning and ending amounts] − Non-controlling interests [average of beginning and ending amounts])
8. Equity ratio = (Net assets − Non-controlling interests) ÷ Total assets
9. Debt-to-equity ratio = Interest-bearing debt ÷ (Net assets − Non-controlling interests)

Company Profile

Dates of establishment	February 1898 Koriyama Kenshi Boseki Co., Ltd. (Koriyama) April 1918 Fukushima Seiren Seishi Co., Ltd. (Fukushima) April 1923 Nitto Boseki Co., Ltd.
Head office	1, Aza Higashi, Gonome, Fukushima-shi, Fukushima, 960-8161, Japan
Headquarters	2-4-1, Kojimachi, Chiyoda-ku, Tokyo, 102-8489, Japan
Paid-in capital	JPY 19.6 billion
Representative and Chief Executive Officer	Hiroyuki Tada
Employees	2,745 (consolidated), 825 (non-consolidated) (As of March 31, 2025)
Business operations	• Manufacture, processing, and sale of glass fiber products • Manufacture, processing, and sale of chemical products and pharmaceutical products • Manufacture, processing, and sale of textile products • Planning, supervision, and contracting for construction • Design, manufacture, and sale of machinery and equipment, etc.

Business Locations (As of April 1, 2025)

Tokyo Headquarters	2-4-1, Kojimachi, Chiyoda-ku, Tokyo, 102-8489, Japan	Tel. +81-3-4582-5111
Osaka Branch	4-2-3, Hiranomachi, Chuo-ku, Osaka-shi, Osaka, 541-0046, Japan (OBIC Midosuji Building)	Tel. +81-6-6208-5015
Nagoya Branch	2-14-21, Nishiki, Naka-ku, Nagoya-shi, Aichi, 460-0003, Japan (Maruyama Nissei Building)	Tel. +81-52-231-5131
Fukushima Enterprise Center	1, Aza Higashi, Gonome, Fukushima-shi, Fukushima, 960-8581, Japan	Tel. +81-24-546-3131
Former Fukushima No. 1 Factory	1, Aza Higashi, Gonome, Fukushima-shi, Fukushima, 960-8581, Japan	Tel. +81-24-546-3131
Former Fukushima No. 2 Factory	20, Ipponsugi, Sakurashimo, Fukushima-shi, Fukushima, 960-2154, Japan	Tel. +81-24-593-1231
Fukuyama Enterprise Center	1, Aza Shiojima, Fukuhara, Fukuyama-cho, Koriyama-shi, Fukushima, 963-8061, Japan	Tel. +81-24-932-6011
	Technology Development Division	Tel. +81-24-932-6110

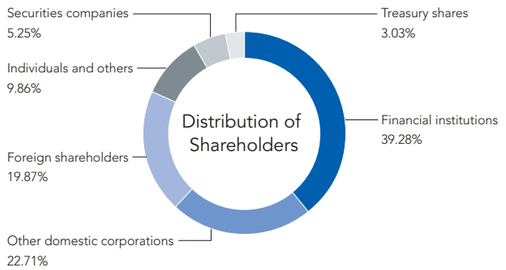
Main Consolidated Subsidiaries (As of April 1, 2025)

Electronic Materials Business Division		
Nitto Glass Fiber Manufacturing Co., Ltd.	Koriyama-shi, Fukushima	Production of glass fiber products
NITTOBO ASIA Glass Fiber Co., Ltd.	Chiayi County, Taiwan	Production and sale of glass fiber products
Baotek Industrial Materials Ltd.	Taoyuan, Taiwan	Production and sale of glass fiber products
Nittobo Taiwan Co., Ltd.	Taoyuan, Taiwan	Sale of glass fiber products
Medical Business Division		
NITTOBO MEDICAL CO., LTD.	Chiyoda-ku, Tokyo	Development, production, and sale of in vitro diagnostic reagents
Frontier Institute Co., Ltd.	Ishikari-shi, Hokkaido	Development, manufacturing, and sale of bioassay kits and consignment of antibody production
Nittobo America Inc.	California, Iowa and Maine, U.S.A.	Production and sale of antisera used for in vitro diagnostic reagents
Kamiya Biomedical Company, LLC	Washington, U.S.A.	Production and sale of in vitro diagnostic reagents
Composite Materials Business Division		
Fuji Fiber Glass Co., Ltd.	Moka-shi, Tochigi	Production, processing, and sale of glass fiber products, etc.
Materials Solution Business Division		
Nitto Glasstex Co., Ltd.	Oizumi-machi, Oura-gun, Gunma	Production and sale of glass fiber products
Nittobo FRP Laboratory Co., Ltd.	Koriyama-shi, Fukushima	Production and sale of glass fiber products, etc.
Nittobo Advantex Co., Ltd.	Itami-shi, Hyogo	Development, production, and sale of interlining products, functional materials, and dishcloths
Bunkyo Seiren Co., Ltd.	Fukui-shi, Fukui	Processing of textile products
Insulation Materials Business Division		
Paramount Glass Manufacturing Co., Ltd.	Sukagawa-shi, Fukushima	Production and sale of glass wool products
Other		
NITTOBO GLOBAL TRADING Co., Ltd.	Chiyoda-ku, Tokyo	Sale of glass fiber products, electronic materials, etc.
Nittobo Techno Co., Ltd.	Koriyama-shi, Fukushima	Design, manufacture, and sale of mechanical facilities; design, construction management, and contracting of civil engineering work and buildings
Nittobo Allied Service Co., Ltd.	Fukushima-shi, Fukushima	Various contracted services and insurance agency
Nittobo Ecology Co., Ltd.	Chiyoda-ku, Tokyo	Various contracted services

Stock Information (As of March 31, 2025)

Stock Status

Authorized number of shares	120,000,000 shares
Number of shares issued	37,723,012 shares
Number of shareholders	10,435



Major Shareholders

Name	Number of shares held (Thousand)	Percentage of shares held (%)
The Master Trust Bank of Japan, Ltd. (Trust account)	3,751	10.25
Custody Bank of Japan, Ltd. (Trust account)	2,471	6.76
Sumitomo Realty & Development Co., Ltd.	2,383	6.52
Nippon Life Insurance Company	1,614	4.41
The Dai-ichi Life Insurance Company, Limited	1,316	3.60
Sumitomo Life Insurance Company	1,082	2.96
AIR WATER INC.	997	2.73
The Toho Bank, Ltd.	905	2.47
Mizuho Bank, Ltd.	800	2.19
Japan Petroleum Exploration Co., Ltd.	660	1.80

Notes: 1. Number of shares held is rounded down to the nearest 1,000 shares.
2. The percentage of shares held is rounded to the nearest third decimal place.
3. The percentage of shares held excludes treasury shares.
Note that treasury shares do not include Company shares held by trust accounts of stock compensation trusts.

Stock-related Administration

Fiscal year	April 1 to March 31
Ordinary General Meeting of Shareholders	Mid-June
Record dates	Voting rights for Ordinary General Meeting of Shareholders: March 31 Year-end dividend: March 31 Interim dividend: September 30
Method of public notice	Public notices are available on the Company's website below: https://www.nittobo.co.jp/koukoku/index.htm (Japanese only) However, if an electronic public notice cannot be posted due to an accident or other unavoidable reason, the public notice will be printed in the Nihon Keizai Shimbun.
Stock listing	Prime Market of the Tokyo Stock Exchange (Securities code: 3110)
Shareholder registry administrator and special account administrator	Mizuho Trust & Banking Co., Ltd. 1-3-3 Marunouchi, Chiyoda-ku, Tokyo, Japan

The Nittobo Group's detailed IR information is available on its official website.

https://www.nittobo.co.jp/eng/ir/index.htm

- Management policy
- Corporate governance
- Financial results-related materials (financial statements and other information)
- News releases
- IR calendar
- Performance and financial information and more

Website | IR information

Message from the Corporate Communication Department

Thank you for reading Nittobo Group Integrated Report 2025.

This year, building on our 100-year history and looking ahead to the next century, we have sought to communicate our determination to achieve medium-to-long-term sustainable growth, centered on the Medium-Term Management Plan (FY2024-2027) that launched last fiscal year.

In addition to expressing our gratitude to all stakeholders, we hope that this report helps deepen your understanding of the Group's initiatives, which are geared toward resolving social issues to realize a sustainable society.

Through emphasis on dialogue with stakeholders, we will continue to engage in communication that helps generate sustainable growth in corporate value.